

Proof It is sufficient to show that $F'(x) = f(x)$. Thus, for part (i) we have

$$\begin{aligned} F'(x) &= D_x \left[-\frac{1}{k} \cos kx + C \right] \\ &= -\frac{1}{k} D_x(\cos kx) + 0 \\ &= -\frac{1}{k} (-\sin kx)(k) = \sin kx. \end{aligned}$$

The proof of (ii) is left to the reader. □

Example 3 Find the most general antiderivative of

(a) $\sin 5x - \cos x$ (b) $\sqrt{x} + 4 \cos 3x$.

Solution

(a) Using (4.35) and (4.36) we obtain

$$-\frac{1}{5} \cos 5x - \sin x + C$$

where C is an arbitrary constant.

(b) By the Power Rule (4.34),

$$\text{an antiderivative of } \sqrt{x} = x^{1/2} \text{ is } \frac{1}{\frac{3}{2}} x^{3/2} = \frac{2}{3} x^{3/2}.$$

By (ii) of (4.35) and (ii) of (4.36),

$$\text{an antiderivative of } 4 \cos 3x \text{ is } 4 \left[\frac{1}{3} \sin 3x \right] = \frac{4}{3} \sin 3x.$$

Thus, the most general antiderivative of $\sqrt{x} + 4 \cos 3x$ is

$$\frac{2}{3} x^{3/2} + \frac{4}{3} \sin 3x + C$$

where C is an arbitrary constant. ■

Equations that involve derivatives of an unknown function f are very common in mathematical applications. Such equations are called **differential equations**. The function f is called a **solution** of the differential equation. To **solve** a differential equation means to find all solutions. Sometimes, in addition to the differential equation we may know certain values of f , called **boundary values**, as illustrated in the next example.

Example 4 Solve the differential equation $f'(x) = 6x^2 + x - 5$ with boundary value $f(0) = 2$.

Solution From our discussion of antiderivatives,

$$f(x) = 2x^3 + \frac{1}{2}x^2 - 5x + C$$

for some number C . Letting $x = 0$ and using the given boundary value, we obtain

$$f(0) = 0 + 0 - 0 + C = 2$$

and hence $C = 2$. Consequently, the solution f of the differential equation with the given boundary value is

$$f(x) = 2x^3 + \frac{1}{2}x^2 - 5x + 2. \quad \blacksquare$$

If a point P is moving rectilinearly, then its position function s is an antiderivative of its velocity function, that is, $s'(t) = v(t)$. Similarly, since $v'(t) = a(t)$, the velocity function is an antiderivative of the acceleration function. If the velocity or acceleration function is known, then given sufficient boundary conditions it is possible to determine the position function. The particular boundary conditions corresponding to $t = 0$ are sometimes called the **initial conditions**.

Example 5 A motorboat moves away from a dock along a straight line, with an acceleration at time t given by $a(t) = 12t - 4$ ft/sec². If, at time $t = 0$, the boat had a velocity of 8 ft/sec and was 15 ft from the dock, find its distance $s(t)$ from the dock at the end of t seconds.

Solution From $v'(t) = 12t - 4$ we obtain, by antidifferentiation,

$$v(t) = 6t^2 - 4t + C$$

for some number C . Substitution of 0 for t and use of the fact that $v(0) = 8$ gives us $8 = 0 - 0 + C = C$. Thus

$$v(t) = 6t^2 - 4t + 8$$

or, equivalently, $s'(t) = 6t^2 - 4t + 8$.

The most general antiderivative of $s'(t)$ is

$$s(t) = 2t^3 - 2t^2 + 8t + D$$

where D is some number. Substitution of 0 for t and use of the fact that $s(0) = 15$ leads to $15 = 0 - 0 + 0 + D = D$ and, consequently, the desired position function is given by

$$s(t) = 2t^3 - 2t^2 + 8t + 15. \quad \blacksquare$$

An object on or near the surface of the Earth is acted upon by a force called **gravity**, which produces a constant acceleration denoted by g . The approximation to g which is employed for most problems is 32 ft/sec² or 980 cm/sec². The use of this important physical constant is illustrated in the following example.

Example 6 A stone is thrown vertically upward from a position 144 ft above the ground with a velocity of 96 ft/sec. If air resistance is neglected, find its distance above the ground after t sec. For what length of time does the stone rise? When, and with what velocity, does it strike the ground?

Solution The motion of the stone may be represented by a point moving rectilinearly on a vertical line with origin at ground level and positive direction upward. The distance above the ground at time t is $s(t)$ and the initial conditions are $s(0) = 144$ and $v(0) = 96$. Since the velocity is decreasing, $v'(t) < 0$; that is, the acceleration is negative. Hence, by the remarks preceding this example,

$$a(t) = -32.$$

Since v is an antiderivative of a ,

$$v(t) = -32t + C,$$

for some number C . Substituting 0 for t and using the fact that $v(0) = 96$ gives us $96 = 0 + C = C$ and, consequently,

$$v(t) = -32t + 96.$$

Since $s'(t) = v(t)$ we obtain, by antidifferentiation,

$$s(t) = -16t^2 + 96t + D$$

for some number D . Letting $t = 0$ and using the fact that $s(0) = 144$ leads to $144 = 0 + 0 + D = D$. It follows that the distance from the ground to the stone at time t is given by

$$s(t) = -16t^2 + 96t + 144.$$

The stone will rise until $v(t) = 0$, that is, until $-32t + 96 = 0$, or $t = 3$. The stone will strike the ground when $s(t) = 0$, that is when $-16t^2 + 96t + 144 = 0$ or, equivalently, $t^2 - 6t - 9 = 0$. Applying the quadratic formula, $t = 3 \pm 3\sqrt{2}$. The solution $3 - 3\sqrt{2}$ is extraneous (Why?) and hence the stone strikes the ground after $3 + 3\sqrt{2}$ sec. The velocity at that time is

$$\begin{aligned} v(3 + 3\sqrt{2}) &= -32(3 + 3\sqrt{2}) + 96 \\ &= -96\sqrt{2} \approx -135.8 \text{ ft/sec.} \end{aligned}$$

■

4.9 Exercises

Find the most general antiderivatives of the functions defined in Exercises 1–28.

1 $f(x) = 9x^2 - 4x + 3$

2 $f(x) = 4x^2 - 8x + 1$

3 $f(x) = 2x^3 - x^2 + 3x - 7$

4 $f(x) = 10x^4 - 6x^3 + 5$

5 $f(x) = \frac{1}{x^3} - \frac{3}{x^2}$

6 $f(x) = \frac{4}{x^7} - \frac{7}{x^4} + x$

- 7 $f(x) = 3\sqrt{x} + (1/\sqrt{x})$
- 8 $f(x) = \sqrt{x^3} - \frac{1}{2}x^{-2} + 5$
- 9 $f(x) = (6/\sqrt[3]{x}) - (\sqrt[3]{x}/6) + 7$
- 10 $f(x) = 3x^5 - \sqrt{x^5}$
- 11 $f(x) = 2x^{5/4} + 6x^{1/4} + 3x^{-4}$
- 12 $f(x) = \left(x - \frac{1}{x}\right)^2$
- 13 $f(x) = 3 \sin 4x$
- 14 $f(x) = 4 \cos \frac{1}{2}x$
- 15 $f(x) = 2x^2 - 3x + 7 \sin 5x$
- 16 $f(x) = (4 + 3x^2 \cos 4x)/x^2$
- 17 $f(x) = 2 \cos 3x - 3 \sin 2x$
- 18 $f(x) = (\sin x + \cos x)^2$ (Hint: $\sin 2\theta = 2 \sin \theta \cos \theta$)
- 19 $f(x) = \sin 4x/\cos 2x$ (Hint: $\sin 2\theta = 2 \sin \theta \cos \theta$)
- 20 $f(x) = \cos 3x \tan 3x$
- 21 $f(x) = (3x - 1)^2$
- 22 $f(x) = (2x - 5)(3x + 1)$
- 23 $f(x) = (8x - 5)/\sqrt[3]{x}$
- 24 $f(x) = (2x^2 - x + 3)/\sqrt{x}$
- 25 $f(x) = \sqrt[5]{32x^4}$
- 26 $f(x) = \sqrt[3]{64x^5}$
- 27 $f(x) = (x^3 - 1)/(x - 1)$
- 28 $f(x) = \frac{x^3 + 3x^2 - 9x - 2}{x - 2}$

Solve the differential equations in Exercises 29–34 subject to the given boundary conditions.

- 29 $f'(x) = 12x^2 - 6x + 1, f(1) = 5$
- 30 $f'(x) = 9x^2 + x - 8, f(-1) = 1$
- 31 $f''(x) = 4x - 1, f'(2) = -2, f(1) = 3$
- 32 $f'''(x) = 6x, f''(0) = 2, f'(0) = -1, f(0) = 4$
- 33 $f'''(t) = \frac{\pi^3}{2} \sin \frac{\pi}{4}t, f(0) = 32, f'(0) = 0, f''(0) = 4 - 2\pi^2$
- 34 $f''(x) = 4 \sin 2x + 16 \cos 4x, f(0) = 6, f'(0) = 1$
- 35 A point moves rectilinearly such that $a(t) = 2 - 6t$. If the initial conditions are $v(0) = -5$ and $s(0) = 4$, find $s(t)$.
- 36 Work Exercise 35 if $a(t) = 3t^2, v(0) = 20$, and $s(0) = 5$.
- 37 A charged particle is moving rectilinearly in a magnetic field such that its velocity (in cm/sec) at time t is given by $v(t) = \frac{1}{2} \sin(3t - \pi/4)$. Show that the motion is simple harmonic.
- 38 The acceleration of a particle that is moving rectilinearly is given by $a(t) = k \cos(\omega t + \phi)$, where k, ω , and ϕ are constants and t is the time in seconds. Show that the motion is simple harmonic.
- 39 A projectile is fired vertically upward from ground level with a velocity of 1600 ft/sec. If air resistance is neglected, find its distance $s(t)$ above ground at time t . What is its maximum height?
- 40 An object is dropped from a height of 1000 ft. Neglecting air resistance, find the distance it falls in t seconds. What is its velocity at the end of 3 sec? When will it strike the ground?
- 41 A stone is thrown directly downward from a height of 96 ft with a velocity of 16 ft/sec. Find (a) its distance above the ground after t seconds; (b) when it will strike the ground; and (c) the velocity at which it strikes the ground.
- 42 The gravitational constant for objects near the surface of the moon is approximately 5.3 ft/sec². If an astronaut throws a stone directly upward with an initial velocity of 60 ft/sec, find its maximum altitude. If, after returning to earth, the astronaut throws the same stone directly upward with the same initial velocity, find the maximum altitude.
- 43 If a projectile is fired vertically upward from a height of s_0 ft above the ground with a velocity of v_0 ft/sec, prove that if air resistance is neglected, its distance $s(t)$ above the ground after t sec is given by $s(t) = -\frac{1}{2}gt^2 + v_0t + s_0$, where g is the gravitational constant.
- 44 A ball rolls down an inclined plane with an acceleration of 2 ft/sec². If the ball is given no initial velocity how far will it roll in t sec? What initial velocity must be given for the ball to roll 100 feet in 5 sec?
- 45 If an automobile starts from rest, what constant acceleration will enable it to travel 500 ft in 10 sec?
- 46 If a car is traveling at a speed of 60 mi/hr, what constant (negative) acceleration will enable it to stop in 9 sec?
- 47 If C and F denote Celsius and Fahrenheit temperature readings, then the rate of change of F with respect to C is given by $dF/dC = \frac{9}{5}$. If $F = 32$ when $C = 0$, use anti-differentiation to obtain a general formula for F in terms of C .
- 48 The rate of change of the temperature T of a solution is given by $dT/dt = \frac{1}{4}t + 10$, where t is the time in minutes and T is measured in degrees Celsius. If the temperature is 5°C at $t = 0$, find a formula which gives T at time t .

- 49 The volume V of a balloon is changing with respect to time t at a rate given by $dV/dt = 3\sqrt{t} + (t/4)$ ft³/sec. If, at $t = 4$, the volume is 20 ft³, express V as a function of t .
- 50 Suppose the slope of the tangent line at any point P on the graph of an equation equals the square of the x -coordinate of P . Find the equation if the graph contains (a) the origin; (b) the point (3, 6); (c) the point $(-1, 1)$. Sketch the graph in each case.
- 51 Suppose F and G are antiderivatives of f and g , respectively. Prove or disprove that (a) FG is an antiderivative of fg , and (b) F/G is an antiderivative of f/g .
- 52 Suppose F is an antiderivative of f . Prove that (a) if F is an even function, then f is odd; (b) if F is an odd function, then f is even.

4.10 Applications to Economics

Calculus has become an important tool for problems which occur in economics. As will be seen in this section, if a function f is used to describe some economic entity, then the adjective *marginal* is employed to specify the derivative f' .

If the cost of producing x units of a certain commodity is denoted by $C(x)$, then C is called a **cost function**. The **average cost** $c(x)$ of one unit is defined by $c(x) = C(x)/x$. In order to use the techniques of calculus, x is regarded as a real number, even though this variable may take on only integer values. We always assume that $x \geq 0$, since the production of a negative number of units has no practical significance. The derivative C' of the cost function is called the **marginal cost function**. If we interpret the derivative as a rate of change (see Section 4.7), then $C'(x)$ is the rate at which the cost changes with respect to the number of units produced. The number $C'(x)$ is referred to as the **marginal cost** associated with the production of x units. Evidently $C'(x) > 0$, since the cost should increase as more units are produced.

If C is a cost function and n is a positive integer, then by Theorem (3.7)

$$C'(n) = \lim_{h \rightarrow 0} \frac{C(n+h) - C(n)}{h}.$$

Hence, if h is small, then

$$C'(n) \approx \frac{C(n+h) - C(n)}{h}.$$

If the number n of units produced is large, economists often let $h = 1$ in the last formula to approximate the marginal cost, obtaining

$$C'(n) \approx C(n+1) - C(n).$$

In this context, the marginal cost associated with the production of n units is (approximately) the cost of producing one more unit.

Some companies find that the cost $C(x)$ of producing x units of a certain commodity is given by a formula such as

$$C(x) = a + bx + dx^2 + kx^3.$$

The constant a represents a fixed overhead charge for items such as rent, heat, light, etc., that are independent of the number of units produced. If the cost of producing one unit were b dollars and no other factors were involved, then the second term bx in the formula would represent the cost of producing x units. If x becomes very large, then the terms dx^2 and kx^3 may significantly affect production costs.

Example 1 An electronics company estimates that the cost (in dollars) of producing x components used in electronic toys is given by

$$C(x) = 200 + 0.05x + 0.0001x^2.$$

Find the cost, the average cost, and the marginal cost of producing
(a) 500 units (b) 1000 units (c) 5000 units.

Solution The average cost of producing x components is given by

$$c(x) = \frac{C(x)}{x} = \frac{200}{x} + 0.05 + 0.0001x.$$

The marginal cost is

$$C'(x) = 0.05 + 0.0002x.$$

We leave it to the reader to verify the entries in the following table, where numbers in the last three columns represent dollars, rounded off to the nearest cent.

x	$C(x)$	$c(x) = \frac{C(x)}{x}$	$C'(x)$
500	250.00	0.50	0.15
1000	350.00	0.35	0.25
5000	2950.00	0.59	1.05

■

The derivative $c'(x)$ of the average cost $c(x)$ is called the **marginal average cost**. Applying the quotient rule to $c(x) = C(x)/x$ we obtain

$$c'(x) = \frac{xC'(x) - C(x)}{x^2}$$

and consequently the average cost will have an extremum if

$$xC'(x) - C(x) = 0$$

that is, if

$$C'(x) = \frac{C(x)}{x} = c(x).$$

Thus, a minimum average cost can occur only when the marginal and average costs are equal.

Example 2 In Example 1, find (a) the number of components that will minimize the average cost, and (b) the minimum average cost.

Solution

(a) From the preceding discussion we must have $C'(x) = c(x)$, that is,

$$0.05 + 0.0002x = \frac{200}{x} + 0.05 + 0.0001x.$$

This equation reduces to

$$0.0001x = \frac{200}{x}$$

or
$$x^2 = \frac{200}{0.0001} = 2,000,000.$$

Consequently,

$$x = \sqrt{2,000,000} \approx 1414.$$

The First or Second Derivative Test can be used to verify that the average cost is minimal for this number of units.

(b) Using the result obtained in part (a), the minimum average cost is

$$c(1414) = \frac{200}{1414} + 0.05 + 0.0001(1414) \approx 0.33. \quad \blacksquare$$

Example 3 A company determines that the cost $C(x)$ of manufacturing x units of a commodity may be approximated by

$$C(x) = 100 + \frac{10}{x} + \frac{x^2}{200}.$$

How many units should be produced in order to minimize the cost?

Solution Since the marginal cost is

$$C'(x) = -\frac{10}{x^2} + \frac{x}{100}$$

the cost function will have an extremum if

$$-\frac{10}{x^2} + \frac{x}{100} = 0$$

or, equivalently,
$$\frac{-1000 + x^3}{100x^2} = 0.$$

Solving for x gives us $x = 10$. It is left to the reader to show that C has an absolute minimum at $x = 10$. ■

A company must consider many factors in order to determine a selling price. In addition to the cost of production and the profit desired, the seller should be aware of the manner in which consumer demand will vary if the price increases. For some commodities there is a constant demand, and changes in price have little effect on sales. For items that are not necessities of life, a price increase will probably lead to a decrease in the number of units sold. Suppose a company knows from past experience that it can sell x units when the price per unit is given by $p(x)$, where p is some type of function. We sometimes say that $p(x)$ is the price per unit when there is a **demand** for x units, and we refer to p as the **demand function** for the commodity. The total income, or total revenue, is the number of units sold times the price per unit, that is, $x \cdot p(x)$. For this reason the function R , defined by

$$R(x) = xp(x)$$

is called the **total revenue function**. The derivatives p' and R' are called the **marginal demand** and **marginal revenue** functions, respectively. They are used to find the rates of change of the demand and total revenue functions with respect to the number of units sold.

If we let $S = p(x)$, then S is the selling price per unit associated with a demand of x units. Since a decrease in S would ordinarily be associated with an increase in x , a demand function p is usually decreasing, that is $p'(x) < 0$ for all x . Demand functions are sometimes defined implicitly by an equation involving S and x , as in the next example.

Example 4 The demand for x units of a certain commodity is related to a selling price of S dollars per unit by means of the equation $2x + S^2 - 12,000 = 0$. Find the demand function, the marginal demand function, the total revenue function, and the marginal revenue function. Find the number of units and the price per unit which yield the maximum revenue. What is the maximum revenue?

Solution Since $S^2 = 12,000 - 2x$ and S is positive, we see that the demand function p is given by

$$S = p(x) = \sqrt{12,000 - 2x}.$$

The domain of p consists of all x such that $12,000 - 2x > 0$, or equivalently, $2x < 12,000$. Thus $0 \leq x < 6000$. The graph of p is sketched in Figure 4.52. In theory, there are no sales if the selling price is $\sqrt{12,000}$, or approximately \$109.54, and when the selling price is close to \$0 the demand is close to 6000.

The marginal demand function p' is given by

$$p'(x) = \frac{-1}{\sqrt{12,000 - 2x}}.$$

The negative sign indicates that a decrease in price is associated with an increase in demand.

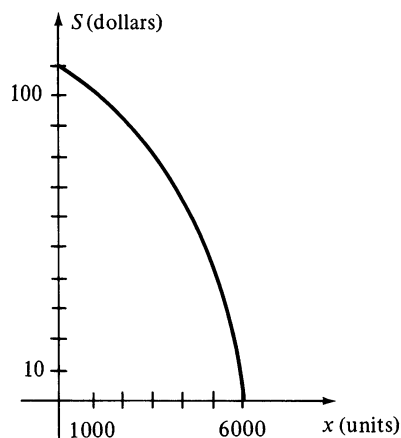


FIGURE 4.52 Graph of demand function

The total revenue function R is given by

$$R(x) = xp(x) = x\sqrt{12,000 - 2x}.$$

Differentiating and simplifying gives us the marginal revenue function R' , where

$$R'(x) = \frac{12,000 - 3x}{\sqrt{12,000 - 2x}}.$$

Thus, $x = 12,000/3 = 4000$ is a critical number for the total revenue function R . Since $R'(x)$ is positive if $0 \leq x < 4000$ and negative if $4000 < x < 6000$, there is a maximum total revenue when 4000 units are produced. This corresponds to a selling price of

$$p(4000) = \sqrt{12,000 - 2(4000)} \approx \$63.25.$$

The maximum revenue, obtained from selling 4000 units at this price is

$$4000(63.25) = \$253,000. \quad \blacksquare$$

If x units of a commodity are sold at a price of $p(x)$ per unit, then the **profit** $P(x)$ is given by

$$P(x) = R(x) - C(x)$$

where R and C are the total revenue and cost functions, respectively. We call P the **profit function** and P' the **marginal profit function**. The critical numbers of P are solutions of the equation

$$P'(x) = R'(x) - C'(x) = 0.$$

This shows that there may be a maximum (or minimum) profit when $R'(x) = C'(x)$, or equivalently when $R'(x) = c(x)$. Thus if P has an extremum then the marginal revenue, the marginal cost, and the average cost are all equal.

Example 5 A furniture company estimates that the weekly cost (in dollars) of manufacturing x hand-finished reproductions of a Colonial period desk is given by

$$C(x) = x^3 - 3x^2 - 80x + 500.$$

Each desk produced is sold for \$2,800. What weekly production rate will maximize the profit? What is the largest possible profit per week?

Solution Since the income obtained from selling x desks is $2800x$, the total revenue function R is given by $R(x) = 2800x$. From the discussion preceding this example, the maximum profit occurs if $C'(x) = R'(x)$, that is, if

$$3x^2 - 6x - 80 = 2800.$$

This equation reduces to

$$x^2 - 2x - 960 = 0, \quad \text{or} \quad (x - 32)(x + 30) = 0$$

and hence either $x = 32$ or $x = -30$. Since the negative solution is extraneous, it suffices to check $x = 32$.

The second derivative of the profit function $P = R - C$ is given by

$$P''(x) = R''(x) - C''(x) = 0 - (6x - 6).$$

Consequently,

$$P''(32) = -6(32) + 6 < 0$$

which means that a maximum profit occurs if 32 desks per week are manufactured. The maximum weekly profit is

$$\begin{aligned} P(32) &= R(32) - C(32) \\ &= 2800(32) - [(32)^3 - 3(32)^2 - 80(32) + 500] \\ &= \$61,964. \end{aligned}$$

■

As a final remark on economic applications, if a marginal function is known, then it is sometimes possible to use antidifferentiation to find the function, as illustrated in the next example.

Example 6 A company that supplies a manufacturer of office equipment finds that the marginal cost (in dollars) associated with the production of x units of a photo-copier component is given by $30 - 0.02x$. If the cost of producing one unit is \$35, find the cost function and the cost of producing 100 units.

Solution If C is the cost function, then the marginal cost is the rate of change of C with respect to x , that is

$$C'(x) = 30 - 0.02x.$$

Antidifferentiation gives us

$$C(x) = 30x - 0.01x^2 + K$$

for some real number K . Letting $x = 1$ and using $C(1) = 35$ we obtain

$$35 = 30 - 0.01 + K$$

and hence $K = 5.01$. Consequently

$$C(x) = 30x - 0.01x^2 + 5.01.$$

In particular, the cost of producing 100 units is

$$C(100) = 3000 - 100 + 5.01 = \$2,905.01.$$

■

4.10 Exercises

In Exercises 1–4, C is the cost function for a certain commodity. Find (a) the cost of producing 100 units; (b) the average and marginal cost functions, and their values at $x = 100$; (c) the minimum average cost; (d) if possible, the value of x that will minimize the cost.

1 $C(x) = 800 + 0.04x + 0.0002x^2$

2 $C(x) = 6400 + 6.5x + 0.003x^2$

3 $C(x) = 250 + 100x + 0.001x^3$

4 $C(x) = 200 + (100/\sqrt{x}) + (\sqrt{x}/1000)$

5 A manufacturer of small motors estimates that the cost (in dollars) of producing x motors per day is given by $C(x) = 25 + 2x + (256/\sqrt{x})$. What daily production will minimize the average cost?

6 A company is pilot-testing the production of a new industrial solvent. It finds that the cost of producing x liters of each pilot run is given by $C(x) = 3 + x + (10/x)$. Compare the marginal cost of producing ten liters with the cost of producing the eleventh liter.

In each of Exercises 7–10 the given equation relates the demand for x units of a certain commodity to a selling price of S dollars per unit. Find the demand function, the marginal demand function, the total revenue function, and the marginal revenue function. If possible, find the number of units and the price per unit which will yield the maximum revenue.

7 $3S + 4x - 800 = 0$

8 $x^2 + S + xS - 400 = 0$

9 $S^2 - 8S - x + 16 = 0$

10 $Sx^2 - 1000x + 144S - 5000 = 0$

In Exercises 11–14, R denotes the total revenue function when there is a demand for x units of a certain commodity. Find (a) the marginal revenue function; (b) the maximum total revenue; and (c) the demand function.

11 $R(x) = 2x\sqrt{400 - x}$

12 $R(x) = 70x - x^2$

13 $R(x) = x(300 - x^2)$

14 $R(x) = 300x - 2x^{3/2}$

In Exercises 15 and 16 the demand and cost functions of a commodity are denoted by p and C , respectively. Find (a) the marginal demand function; (b) the total revenue function; (c) the profit function; (d) the marginal profit function; (e) the maximum profit; and (f) the marginal cost when the demand is 10 units.

15 $p(x) = 50 - (x/10)$; $C(x) = 10 + 2x$

16 $p(x) = 80 - \sqrt{x - 1}$; $C(x) = 75x + 2\sqrt{x - 1}$

17 A travel agency estimates that to sell x “package deal” vacations, the price per vacation should be $1800 - 2x$ dollars, where $1 \leq x \leq 100$. If the cost to the agency for x vacations is $1000 + x + 0.01x^2$ dollars, find (a) the total revenue function; (b) the profit function; (c) the number of vacations that will maximize the profit; and (d) the maximum profit.

18 A manufacturer determines that x units of a product will be sold if the selling price is $400 - 0.05x$ dollars for each unit. If the production cost for x units is $500 + 10x$, find (a) the total revenue function; (b) the profit function; (c) the number of units that maximize the profit; and (d) the price per unit when the marginal revenue is 300.

19 A kitchen specialty company determines that the cost of manufacturing and packaging x pepper grinders per day is $500 + 0.02x + 0.001x^2$ dollars. If each item is sold for \$8.00, what rate of production will maximize the profit? What is the maximum daily profit?

20 A company that conducts bus tours found that when the price was \$9.00 per person, it averaged 1000 customers per week. When it reduced the price to \$7.00 per person, the average number of customers increased to 1500 per week. Assuming that the demand function is linear, what price should be charged to obtain the greatest weekly revenue?

21 When a company sold a certain commodity at \$50 per unit, there was a demand for 1000 units per week. After the price rose to \$70, the demand dropped to 800 units per week. Assuming that the demand function p is linear, find p and the total revenue function.

22 If a demand function p is defined by $p(x) = ax^2 + b$ where $a < 0$ and $b > 0$, what value of x will maximize the total revenue?

23 Prove that if the demand function of a commodity is linear and decreasing for all $x > 0$, then so is the marginal revenue function.

24 Show that the marginal revenue of a commodity can be found by adding to the selling price the product of the number of units sold and the marginal demand.

25 A sportswear manufacturer determines that the marginal cost of producing x sweatsuits is given in dollars by $20 - 0.015x$. If the cost of producing one suit is \$25, find the cost function and the cost of producing 50 suits.

26 If the marginal cost function is given by $2/(x + 6)^{1/3}$, and if the cost of producing two units is \$20, find the cost function and the cost of producing 120 units.

- 27 If, in a certain company, the marginal revenue is given by $x^2 - 6x + 15$, find the total revenue function, and the marginal demand function.

- 28 Work Exercise 27 if the marginal revenue is given by $4/(x + 2)^{3/2}$.

4.11 Review: Concepts

Define or discuss each of the following.

- 1 Local maximum or local minimum of a function
- 2 Absolute maximum or absolute minimum of a function
- 3 Critical numbers of a function
- 4 Rolle's Theorem
- 5 The Mean Value Theorem
- 6 The First Derivative Test
- 7 Endpoint extrema of a function
- 8 Upward or downward concavity
- 9 Tests for concavity
- 10 Point of inflection
- 11 The Second Derivative Test

- 12 $\lim_{x \rightarrow \infty} f(x) = L$; $\lim_{x \rightarrow -\infty} f(x) = L$
- 13 $\lim_{x \rightarrow a} f(x) = \infty$; $\lim_{x \rightarrow a} f(x) = -\infty$
- 14 Vertical and horizontal asymptotes
- 15 The derivative as a rate of change
- 16 Velocity and acceleration in rectilinear motion
- 17 Simple harmonic motion
- 18 Related rates
- 19 Antiderivative of a function
- 20 The power rule for antidifferentiation
- 21 Differential equation
- 22 Applications to economics

Review: Exercises

- 1 Find equations of the tangent and normal lines to the graph of the equation $6x^2 - 2xy + y^3 = 9$ at the point $P(2, -3)$.
- 2 Find equations of the tangent lines to the graph of the equation $x^2 + y^2 = 1$ that contain the point $P(2, 1)$.

In Exercises 3 and 4, find the x -coordinates of points on the graph of the given equation where the tangent line is either horizontal or vertical.

- 3 $y = (x^2 + 3x + 2)^{1/3}$
- 4 $x^2 - 2xy + y^2 - 4 = 0$
- 5 If $f(x) = x^3 + x^2 + x + 1$, find a number c that satisfies the conclusion of the Mean Value Theorem on the interval $[0, 4]$.

In Exercises 6–8 find the local extrema of f by means of the First Derivative Test. Describe the intervals in which f is increasing or decreasing and sketch the graph of f .

- 6 $f(x) = -x^3 + 4x^2 - 3x$
- 7 $f(x) = 1/(1 + x^2)$
- 8 $f(x) = (4 - x^2)x^{1/3}$

In Exercises 9–11 find the local extrema of f by means of the Second Derivative Test. Discuss concavity, find x -coordinates of points of inflection, and sketch the graph of f .

- 9 $f(x) = -x^3 + 4x^2 - 3x$
- 10 $f(x) = 1/(1 + x^2)$
- 11 $f(x) = 40x^3 - x^6$
- 12 If $f(x) = 2 \sin x - \cos 2x$ find equations of the tangent and normal lines to the graph of f at the point $(\pi/6, \frac{1}{2})$.
- 13 If $f(x) = 2 \sin x - \cos 2x$ find the local extrema and sketch the graph of f for $0 \leq x \leq 4\pi$.
- 14 The position function of a point moving rectilinearly is given by $s(t) = (t^2 + 3t + 1)/(t^2 + 1)$. Find the velocity and acceleration at time t and describe the motion of the point during the time interval $[-2, 2]$.
- 15 The position of a moving point on a coordinate line is given by

$$s(t) = a \sin(kt + m) + b \cos(kt + m)$$

where a, b, k , and m are constants. Prove that the magnitude of the acceleration is directly proportional to the distance from the origin.

- 16 A ladder 12 ft long leans against a house. If the lower end slides along level ground at a rate of 3 ft/sec, how fast is the angle between the ladder and the ground changing when the lower end is 6 ft from the house?
- 17 A stone is thrown directly downward from a height of 900 ft with a velocity of 30 ft/sec. Find its distance above ground after t seconds. What is its velocity after 5 sec? When will it strike the ground?

In Exercises 18–23 find the most general antiderivative of f .

- 18 $f(x) = \frac{8x^2 - 4x + 5}{x^4}$
- 19 $f(x) = 3x^5 + 2x^3 - x$
- 20 $f(x) = 100$
- 21 $f(x) = x^{3/5}(2x - \sqrt{x})$
- 22 $f(x) = (2x + 1)^3$
- 23 $f(t) = 3 \cos 2\pi t - 5 \sin 4\pi t$
- 24 Solve the differential equation $f''(x) = x^{1/3} - 5$ if $f'(1) = 2$ and $f(1) = -8$.
- 25 A man wishes to put a fence around a rectangular field and then subdivide this field into three smaller rectangular plots by placing two fences parallel to one of the sides. If he can afford only 1000 yd of fencing, what dimensions will give him the maximum area?
- 26 A V-shaped water gutter is to be constructed from two rectangular sheets of metal 10 in. wide. Find the angle between the sheets that will maximize the carrying capacity of the gutter.
- 27 Find the altitude of the right circular cylinder of maximum curved surface area that can be inscribed in a sphere of radius a .
- 28 A wire 5 ft long is to be cut into two pieces. One of the pieces is to be bent into the shape of a circle and the other into the shape of a square. Where should the wire be cut in order that the sum of the areas of the circle and square is (a) a maximum; (b) a minimum?
- 29 The interior of a half-mile race track consists of a rectangle with semicircles at two opposite ends. Find the dimensions which will maximize the area of the rectangle.
- 30 A rectangle has its vertices on the x -axis, the y -axis, the origin, and the graph of $y = 4 - x^2$. Of all such rectangles, find the dimensions of the one with maximum area.
- 31 A water tank has the shape of a right circular cone of altitude 12 ft and base radius 4 ft, with vertex at the bottom of the tank. If water is being taken out of the tank at a rate of 10 ft³/min, how fast is the water level falling when the depth is 5 ft?

- 32 The ends of a horizontal water trough 10 ft long are isosceles trapezoids with lower base 3 ft, upper base 5 ft, and altitude 2 ft. If the water level is rising at a rate of $\frac{1}{4}$ in./min when the depth is 1 ft, how fast is water entering the trough?
- 33 Two cars are approaching the same intersection along roads that run at right angles to each other. Car A is traveling at 20 mi/hr and car B is traveling at 40 mi/hr. If, at a certain instant, A is $\frac{1}{4}$ mile from the intersection and B is $\frac{1}{2}$ mile from the intersection, find the rate at which they are approaching one another at that instant.
- 34 A point $P(x, y)$ moves on the graph of $y^2 = 2x^3$ such that $dy/dt = x$, where t is time. Find dx/dt at the point (2, 4).
- 35 Boyle's Law states that $p v = c$, where p is pressure, v is volume, and c is a constant. Find a formula for the rate of change of p with respect to v .
- 36 A railroad bridge is 20 ft above, and at right angles to, a river. A man in a train traveling 60 mi/hr passes over the center of the bridge at the same instant that a man in a motor boat traveling 20 mi/hr passes under the center of the bridge (see figure). How fast are the two men separating 10 seconds later?

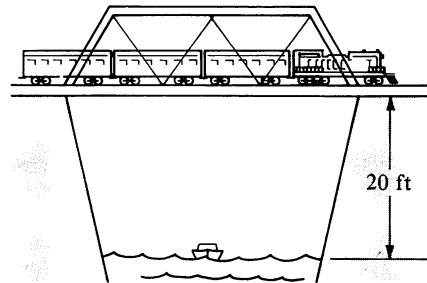


FIGURE FOR EXERCISE 36

Find each of the limits in Exercises 37–44, if it exists.

- 37 $\lim_{x \rightarrow -\infty} \frac{(2x - 5)(3x + 1)}{(x + 7)(4x - 9)}$
- 38 $\lim_{x \rightarrow \infty} \frac{2x + 11}{\sqrt{x + 1}}$
- 39 $\lim_{x \rightarrow -\infty} \frac{6 - 7x}{(3 + 2x)^4}$
- 40 $\lim_{x \rightarrow -3} \frac{\sqrt[3]{x + 3}}{\sqrt{x^3 + 27}}$
- 41 $\lim_{x \rightarrow 2/3^+} \frac{x^2}{4 - 9x^2}$
- 42 $\lim_{x \rightarrow 3/5^-} \frac{1}{5x - 3}$
- 43 $\lim_{x \rightarrow 0^+} \left(\sqrt{x} - \frac{1}{\sqrt{x}} \right)$
- 44 $\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{(x - 1)^2}}$

Find horizontal and vertical asymptotes and sketch the graph of f in Exercises 45 and 46.

- 45 $f(x) = \frac{3x^2}{9x^2 - 25}$
- 46 $f(x) = \frac{x^2}{(x - 1)^2}$

5 *The Definite Integral*

Calculus consists of two main parts, *differential calculus* and *integral calculus*. Differential calculus is based upon the derivative. In this chapter we define the concept which is the basis for integral calculus: the *definite*

integral. One of the most important results we shall discuss is the *Fundamental Theorem of Calculus*. This theorem demonstrates that differential and integral calculus are very closely related.

5.1 *Area*

In our development of the definite integral we shall employ sums of many numbers. To express such sums compactly, it is convenient to use **summation notation**. To illustrate, given a collection of numbers $\{a_1, a_2, \dots, a_n\}$, the symbol $\sum_{i=1}^n a_i$ represents their sum, that is,

$$(5.1) \quad \sum_{i=1}^n a_i = a_1 + a_2 + a_3 + \cdots + a_n.$$

In (5.1), the Greek capital letter Σ (sigma) indicates a sum, and the symbol a_i represents the i th term. The letter i is called the **index of summation** or the **summation variable**, and the numbers 1 and n indicate the extreme values of the summation variable.

Example 1 Find $\sum_{i=1}^4 i^2(i-3)$.

Solution In this case, $a_i = i^2(i-3)$. To find the indicated sum we merely substitute, in succession, the integers 1, 2, 3, and 4 for i and add the resulting terms. Thus,

$$\begin{aligned}\sum_{i=1}^4 i^2(i-3) &= 1^2(1-3) + 2^2(2-3) + 3^2(3-3) + 4^2(4-3) \\ &= (-2) + (-4) + 0 + 16 = 10. \quad \blacksquare\end{aligned}$$

The letter used for the summation variable is arbitrary. The sum in Example 1 can be written

$$\sum_{i=1}^4 i^2(i-3) = \sum_{k=1}^4 k^2(k-3) = \sum_{j=1}^4 j^2(j-3)$$

or in many other ways.

If $a_i = c$ for each i then, for example,

$$\begin{aligned}\sum_{i=1}^2 a_i &= a_1 + a_2 = c + c = 2c = \sum_{i=1}^2 c \\ \sum_{i=1}^3 a_i &= a_1 + a_2 + a_3 = c + c + c = 3c = \sum_{i=1}^3 c.\end{aligned}$$

In general, the following result is true for every positive integer n .

$$(5.2) \quad \sum_{i=1}^n c = nc$$

for every real number c .

The domain of the summation variable does not have to begin at 1. For example, the following is self-explanatory:

$$\sum_{i=4}^8 a_i = a_4 + a_5 + a_6 + a_7 + a_8.$$

Example 2 Find $\sum_{i=0}^3 \frac{2^i}{(i+1)}$.

$$\begin{aligned}\text{Solution} \quad \sum_{i=0}^3 \frac{2^i}{(i+1)} &= \frac{2^0}{(0+1)} + \frac{2^1}{(1+1)} + \frac{2^2}{(2+1)} + \frac{2^3}{(3+1)} \\ &= 1 + 1 + \frac{4}{3} + 2 = \frac{16}{3}. \quad \blacksquare\end{aligned}$$

Theorem (5.3)

If n is any positive integer and $\{a_1, a_2, \dots, a_n\}$ and $\{b_1, b_2, \dots, b_n\}$ are sets of numbers, then

$$(i) \quad \sum_{i=1}^n (a_i + b_i) = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i;$$

$$(ii) \quad \sum_{i=1}^n ca_i = c \left(\sum_{i=1}^n a_i \right), \text{ for any number } c;$$

$$(iii) \quad \sum_{i=1}^n (a_i - b_i) = \sum_{i=1}^n a_i - \sum_{i=1}^n b_i.$$

Proof To prove formula (i) we begin by writing

$$\sum_{i=1}^n (a_i + b_i) = (a_1 + b_1) + (a_2 + b_2) + (a_3 + b_3) + \cdots + (a_n + b_n).$$

The terms on the right may be rearranged to produce

$$\sum_{i=1}^n (a_i + b_i) = (a_1 + a_2 + a_3 + \cdots + a_n) + (b_1 + b_2 + b_3 + \cdots + b_n).$$

Expressing the right side in summation notation gives us formula (i).

For formula (ii) we have

$$\begin{aligned} \sum_{i=1}^n (ca_i) &= ca_1 + ca_2 + ca_3 + \cdots + ca_n \\ &= c(a_1 + a_2 + a_3 + \cdots + a_n) = c \left(\sum_{i=1}^n a_i \right). \end{aligned}$$

The proof of (iii) is left as an exercise. □

Before stating the definition of the definite integral, it will be instructive to consider the area of a certain region in a plane. Physical examples could be used; however, we prefer to postpone them until the next chapter. It is important to remember that the discussion of area in this section is *not* to be considered as the definition of the definite integral. It is included only to help motivate the work in Section 5.2 in the same way that slopes of tangent lines and velocities were used to motivate the definition of derivative.

It is easy to calculate the area of a plane region bounded by straight lines. For example, the area of a rectangle is the product of its length and width. The area of a triangle is one-half the product of the altitude and base. The area of any polygon can be found by subdividing it into triangles. To find areas of more complicated regions, whose boundaries involve graphs of functions, it is necessary to introduce a limiting process and then use methods of calculus. In particular, let us consider a region S in a coordinate plane, bounded by vertical lines with x -intercepts a and b , by the x -axis, and by the

graph of a function f , which is continuous and nonnegative on the closed interval $[a, b]$. A region of this type is illustrated in Figure 5.1. Since $f(x) \geq 0$ for every x in $[a, b]$, no part of the graph lies below the x -axis. For convenience we shall refer to S as **the region under the graph of f from a to b** . Our objective is to define the area of S .

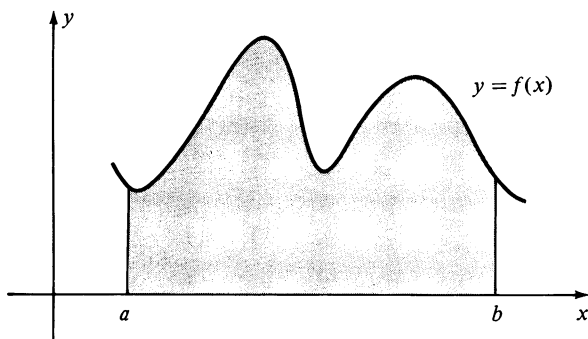


FIGURE 5.1

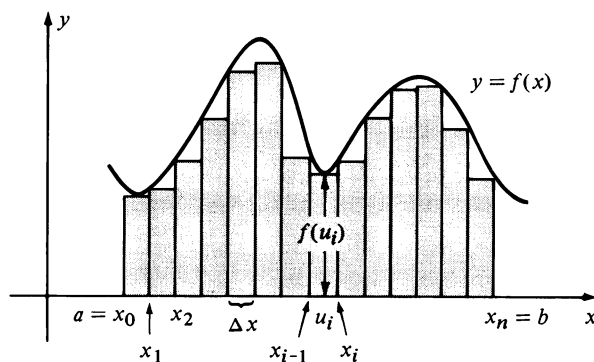


FIGURE 5.2

If n is any positive integer, let us begin by dividing the interval $[a, b]$ into n subintervals, all having the same length $(b - a)/n$. This can be accomplished by choosing numbers $x_0, x_1, x_2, \dots, x_n$ where $a = x_0, b = x_n$, and

$$x_i - x_{i-1} = \frac{b - a}{n}$$

for $i = 1, 2, \dots, n$. If the length $(b - a)/n$ of each subinterval is denoted by Δx , then for each i we have

$$\Delta x = x_i - x_{i-1}, \quad \text{and} \quad x_i = x_{i-1} + \Delta x$$

as illustrated in Figure 5.2.

Note that

$$\begin{aligned} x_0 &= a, & x_1 &= a + \Delta x, & x_2 &= a + 2\Delta x, & \dots, \\ x_i &= a + i\Delta x, & \dots, & & x_n &= a + n\Delta x = b. \end{aligned}$$

Since f is continuous on each subinterval $[x_{i-1}, x_i]$, it follows from Theorem (4.3) that f takes on a minimum value at some number u_i in $[x_{i-1}, x_i]$. For each i , let us construct a rectangle of width $\Delta x = x_i - x_{i-1}$ and length equal to the minimum distance $f(u_i)$ from the x -axis to the graph of f , as shown in Figure 5.2. The area of the i th rectangle is $f(u_i) \Delta x$. The boundary of the region formed by the totality of these rectangles is called the **inscribed rectangular polygon** associated with the subdivision of $[a, b]$ into n subintervals. The area of this inscribed polygon is the sum of the areas of the n rectangles, that is,

$$f(u_1) \Delta x + f(u_2) \Delta x + \dots + f(u_n) \Delta x.$$

Using summation notation we may write:

$$\text{Area of inscribed rectangular polygon} = \sum_{i=1}^n f(u_i) \Delta x$$

where $f(u_i)$ is the minimum value of f on $[x_{i-1}, x_i]$.

Referring to Figure 5.2, we see that if n is very large or, equivalently, if Δx is very small, then the sum of the rectangular areas appears to be close to what we wish to consider as the area of the region S . Indeed, reasoning intuitively, if there exists a number A that has the property that the sum $\sum_{i=1}^n f(u_i) \Delta x$ gets closer and closer to A as Δx gets closer and closer to 0 (but $\Delta x \neq 0$), then we shall call A the **area** of S and write

$$A = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n f(u_i) \Delta x.$$

The meaning of this “limit of a sum” is not the same as that for limit of a function, introduced in Chapter 2. To eliminate the hazy phrase “closer and closer” and arrive at a satisfactory definition of A , let us take a slightly different point of view. If A denotes the area of S , then the difference

$$A - \sum_{i=1}^n f(u_i) \Delta x$$

is the area of the unshaded portion in Figure 5.2 that lies under the graph of f and over the inscribed rectangular polygon. This number may be thought of as the error involved in using the area of the inscribed rectangular polygon to approximate A . If we have the proper notion of area, then we should be able to make this error arbitrarily small by choosing the width Δx of the rectangles sufficiently small. This is the motivation for the following definition of the area A of S , where the notation is the same as that used in the preceding discussion.

Definition (5.4)

Let f be continuous and nonnegative on $[a, b]$. Let A be a real number and let $f(u_i)$ be the minimum value of f on $[x_{i-1}, x_i]$. The statement

$$A = \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n f(u_i) \Delta x$$

means that for every $\varepsilon > 0$ there corresponds a $\delta > 0$ such that if $0 < \Delta x < \delta$, then

$$A - \sum_{i=1}^n f(u_i) \Delta x < \varepsilon.$$

If A is the indicated limit and we let $\varepsilon = 10^{-9}$, then Definition (5.4) states that by using sufficiently thin rectangles, the difference between A and the area of the inscribed polygon is less than one-billionth of a square

unit! Similarly, if $\varepsilon = 10^{-12}$ we can make this difference less than one-trillionth of a square unit. In general, the difference can be made less than any preassigned ε .

If f is continuous on $[a, b]$, then, as it is shown in more advanced texts, a number A satisfying Definition (5.4) actually exists. We shall call A **the area under the graph of f from a to b** .

The area A may also be obtained by means of **circumscribed rectangular polygons** of the type illustrated in Figure 5.3. In this case we select the number v_i in each interval $[x_{i-1}, x_i]$, such that $f(v_i)$ is the maximum value of f on $[x_{i-1}, x_i]$.

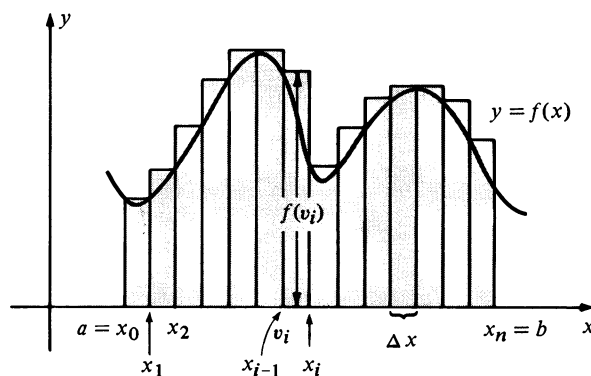


FIGURE 5.3

We may then write:

$$\text{Area of circumscribed rectangular polygon} = \sum_{i=1}^n f(v_i) \Delta x.$$

The limit of this sum as $\Delta x \rightarrow 0$ is defined as in (5.4), where the only change is that we use

$$\sum_{i=1}^n f(v_i) \Delta x - A < \varepsilon$$

in the definition, since we want this difference to be nonnegative. It can be proved that the same number A is obtained using either inscribed or circumscribed rectangles.

The following formulas will be useful in some illustrations of Definition (5.4).

$$(5.5) \quad \begin{aligned} \text{(i)} \quad & \sum_{i=1}^n i = 1 + 2 + \cdots + n = \frac{n(n+1)}{2} \\ \text{(ii)} \quad & \sum_{i=1}^n i^2 = 1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6} \\ \text{(iii)} \quad & \sum_{i=1}^n i^3 = 1^3 + 2^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2} \right]^2 \end{aligned}$$

These may be proved by means of mathematical induction (see Appendix I).

The next two examples provide specific illustrations of how summation properties may be used in conjunction with Definition (5.4) to find the areas of certain regions in a coordinate plane.

Example 3 If $f(x) = 16 - x^2$, find the area of the region under the graph of f from 0 to 3.

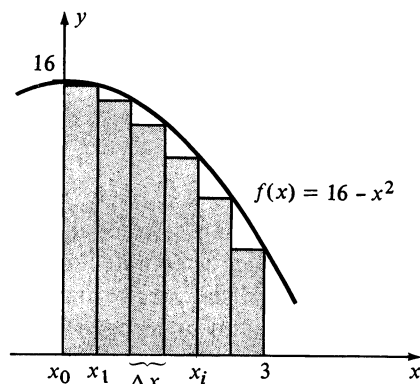


FIGURE 5.4

Solution The region is illustrated in Figure 5.4 where for clarity we have used different scales on the x - and y -axes. If the interval $[0, 3]$ is divided into n equal subintervals, then the length Δx of a typical subinterval is $3/n$. Employing the notation used in Figure 5.2, with $a = 0$ and $b = 3$,

$$x_0 = 0, x_1 = \Delta x, x_2 = 2(\Delta x), \dots, x_i = i(\Delta x), \dots, x_n = n(\Delta x) = 3.$$

Using the fact that $\Delta x = 3/n$ we may write

$$x_i = i(\Delta x) = i\left(\frac{3}{n}\right) = \frac{3i}{n}.$$

Since f is decreasing on $[0, 3]$, the number u_i in $[x_{i-1}, x_i]$ at which f takes on its minimum value is always the right-hand endpoint x_i of the subinterval, that is, $u_i = x_i = 3i/n$. Since

$$f(u_i) = f\left(\frac{3i}{n}\right) = 16 - \left(\frac{3i}{n}\right)^2 = 16 - \frac{9i^2}{n^2},$$

the summation in Definition (5.4) may be written

$$\begin{aligned} \sum_{i=1}^n f(u_i) \Delta x &= \sum_{i=1}^n \left(16 - \frac{9i^2}{n^2}\right) \left(\frac{3}{n}\right) \\ &= \sum_{i=1}^n \left(\frac{48}{n} - \frac{27i^2}{n^3}\right). \end{aligned}$$

Using Theorem (5.3) and (5.2), the last sum may be simplified as follows:

$$\sum_{i=1}^n \frac{48}{n} - \sum_{i=1}^n \frac{27i^2}{n^3} = \left(\frac{48}{n}\right)n - \frac{27}{n^3} \sum_{i=1}^n i^2.$$

Next, applying (ii) of (5.5), we obtain

$$\begin{aligned} \sum_{i=1}^n f(u_i) \Delta x &= 48 - \frac{27}{n^3} \left[\frac{n(n+1)(2n+1)}{6} \right] \\ &= 48 - \frac{9}{2n^3} [2n^3 + 3n^2 + n]. \end{aligned}$$

In order to find the area, we must now let Δx approach 0. Since $\Delta x = (b - a)/n$, this can be accomplished by letting n increase without bound. Although our discussion of limits involving infinity in Section 4.5 was concerned with a real variable x , a similar discussion can be given for the

integer variable n . Assuming that this is true, and that we can replace $\Delta x \rightarrow 0$ by $n \rightarrow \infty$, we have

$$\begin{aligned} \lim_{\Delta x \rightarrow 0} \sum_{i=1}^n f(u_i) \Delta x &= \lim_{n \rightarrow \infty} \left\{ 48 - \frac{9}{2n^3} [2n^3 + 3n^2 + n] \right\} \\ &= \lim_{n \rightarrow \infty} 48 - \frac{9}{2} \lim_{n \rightarrow \infty} \left[\frac{2n^3 + 3n^2 + n}{n^3} \right] \\ &= 48 - \frac{9}{2} \lim_{n \rightarrow \infty} \left[2 + \frac{3}{n} + \frac{1}{n^2} \right] \\ &= 48 - \frac{9}{2} [2 + 0 + 0] = 48 - 9 = 39. \quad \blacksquare \end{aligned}$$

Because of the assumptions we made, the preceding solution is not completely rigorous. Indeed, one reason for introducing the definite integral is to enable us to solve problems of this type in a simple and precise manner.

The area in the preceding example may also be found by using circumscribed rectangular polygons. In this case we select, in each subinterval $[x_{i-1}, x_i]$, the number $v_i = (i-1)(3/n)$ at which f takes on its maximum value.

The next example illustrates the use of circumscribed rectangles in finding an area.

Example 4 If $f(x) = x^3$, find the area under the graph of f from 0 to b , where $b > 0$.

Solution If we subdivide the interval $[0, b]$ into n equal parts, then as Figure 5.5 illustrates, we obtain a typical circumscribed rectangular polygon where, as in Example 3,

$$\Delta x = \frac{b}{n} \quad \text{and} \quad x_i = i(\Delta x).$$

For clarity different scales have been used on the x - and y -axes.

Since f is an increasing function, the maximum value $f(v_i)$ in the interval $[x_{i-1}, x_i]$ occurs at the right-hand endpoint, that is,

$$v_i = x_i = i(\Delta x) = i\left(\frac{b}{n}\right) = \frac{bi}{n}.$$

The sum of the areas of the circumscribed rectangles is

$$\begin{aligned} \sum_{i=1}^n f(v_i) \Delta x &= \sum_{i=1}^n \left(\frac{bi}{n}\right)^3 \left(\frac{b}{n}\right) = \sum_{i=1}^n \frac{b^4}{n^4} i^3 \\ &= \frac{b^4}{n^4} \sum_{i=1}^n i^3 = \frac{b^4}{n^4} \left[\frac{n(n+1)}{2} \right]^2 \end{aligned}$$

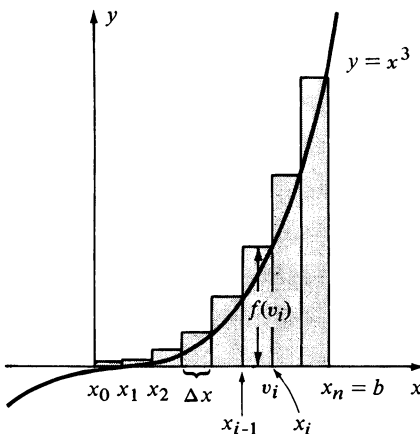


FIGURE 5.5

where the final step follows from (iii) of (5.5). The reader may now verify that

$$\sum_{i=1}^n f(v_i) \Delta x = \frac{b^4}{4} \left(\frac{n^4 + 2n^3 + n^2}{n^4} \right) = \frac{b^4}{4} \left(1 + \frac{2}{n} + \frac{1}{n^2} \right).$$

If we let Δx approach 0, then n increases without bound and the expression in parentheses approaches 1. It follows that the area under the graph is

$$\lim_{\Delta x \rightarrow 0} \sum_{i=1}^n f(v_i) \Delta x = \frac{b^4}{4}.$$

■

5.1 Exercises

Determine the sums in Exercises 1–10.

- 1 $\sum_{i=1}^5 (3i - 10)$
- 2 $\sum_{i=1}^6 (9 - 2i)$
- 3 $\sum_{j=1}^4 (j^2 + 1)$
- 4 $\sum_{n=1}^{10} [1 + (-1)^n]$
- 5 $\sum_{k=0}^5 k(k - 1)$
- 6 $\sum_{k=0}^4 (k - 2)(k - 3)$
- 7 $\sum_{i=1}^8 2^i$
- 8 $\sum_{s=1}^6 (5/s)$
- 9 $\sum_{i=1}^{50} 10$
- 10 $\sum_{k=1}^{1000} 2$
- 11 Prove (iii) of Theorem (5.3).
- 12 Extend (i) of Theorem (5.3) to $\sum_{i=1}^n (a_i + b_i + c_i)$.
- 13 Find $\sum_{i=1}^n (i^2 + 3i + 5)$. (*Hint*: Write the sum as $\sum_{i=1}^n i^2 + 3 \sum_{i=1}^n i + \sum_{i=1}^n 5$ and employ (5.5) and (5.2).)
- 14 Find $\sum_{i=1}^n (3i^2 - 2i + 1)$.
- 15 Find $\sum_{k=1}^n (2k - 3)^2$.
- 16 Find $\sum_{k=1}^n (k^3 + 2k^2 - k + 4)$.

In each case sketch the graph and typical rectangles, labeling the drawing as in Figures 5.4 and 5.5.

- 17 $f(x) = 2x + 3$; $a = 0, b = 4$
- 18 $f(x) = 8 - 3x$; $a = 0, b = 2$
- 19 $f(x) = x^2$; $a = 0, b = 5$
- 20 $f(x) = x^2 + 2$; $a = 1, b = 3$
- 21 $f(x) = 3x^2 + 5$; $a = 1, b = 4$ (*Hint*: $x_i = 1 + (3i/n)$.)
- 22 $f(x) = 7$; $a = -2, b = 6$
- 23 $f(x) = 9 - x^2$; $a = 0, b = 3$
- 24 $f(x) = 4x^2 + 3x + 2$; $a = 1, b = 5$
- 25 $f(x) = x^3 + 1$; $a = 1, b = 2$
- 26 $f(x) = 4x + x^3$; $a = 0, b = 2$
- 27 Use Definition (5.4) to prove that the area of a right triangle of altitude h and base b is $\frac{1}{2}bh$. (*Hint*: Consider the area under the graph of $f(x) = (h/b)x$ from 0 to b .)
- 28 If $f(x) = px^2 + qx + r$ and $f(x) \geq 0$ for all x , prove that the area under the graph of f from 0 to b is $p(b^3/3) + q(b^2/2) + rb$.
- 29 Use mathematical induction to prove (5.5).
- 30 Prove (i) of (5.5) by writing $S = 1 + 2 + \cdots + n$
 $S = n + (n - 1) + \cdots + 1$
 and then adding corresponding sides of these equations.

In Exercises 17–26 find the area under the graph of f from a to b using (a) inscribed rectangles; (b) circumscribed rectangles.

5.2 Definition of Definite Integral

Limiting processes similar to the one for areas used in the preceding section arise frequently in mathematics and its applications. The situations that occur often lead to limits of the form

$$\lim_{\Delta x \rightarrow 0} \sum_{i=1}^n f(w_i) \Delta x.$$

A special instance of this limit appeared in Definition (5.4); however, in the general case, there are several major differences from our work with areas. In our discussion of area in the previous section we made the following assumptions:

- 1 The function f is continuous on a closed interval $[a, b]$.
- 2 $f(x)$ is nonnegative for all x in $[a, b]$.
- 3 All the subintervals $[x_{i-1}, x_i]$ determined by the subdivision of $[a, b]$ have the same length Δx .
- 4 The numbers w_i are chosen such that $f(w_i)$ is always the minimum (or maximum) value of f on $[x_{i-1}, x_i]$.

These four conditions are not always present in applied problems. For this reason it is necessary to allow the following changes in 1–4.

- 1' The function f may be discontinuous at some numbers in $[a, b]$.
- 2' $f(x)$ may be negative for some x in $[a, b]$.
- 3' The lengths of the subintervals $[x_{i-1}, x_i]$ may be different.
- 4' The number w_i is any number in $[x_{i-1}, x_i]$.

We shall begin by introducing some new terminology and notation. A **partition** P of a closed interval $[a, b]$ is any decomposition of $[a, b]$ into subintervals of the form

$$[x_0, x_1], [x_1, x_2], [x_2, x_3], \dots, [x_{n-1}, x_n]$$

where n is a positive integer and the x_i are numbers such that

$$a = x_0 < x_1 < x_2 < x_3 < \dots < x_{n-1} < x_n = b.$$

The length of the i th subinterval $[x_{i-1}, x_i]$ will be denoted by Δx_i , that is,

$$\Delta x_i = x_i - x_{i-1}.$$

A typical partition of $[a, b]$ is illustrated in Figure 5.6. The largest of the numbers $\Delta x_1, \Delta x_2, \dots, \Delta x_n$ is called the **norm** of the partition P and is denoted by $\|P\|$.

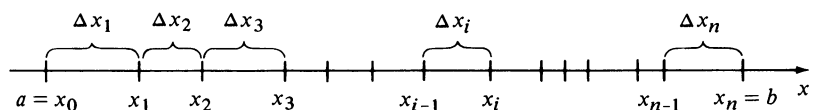


FIGURE 5.6 A partition of $[a, b]$

The following concept, named after the mathematician G. F. B. Riemann (1826–1866), is fundamental for the definition of the definite integral.

Definition (5.6)

Let f be a function that is defined on a closed interval $[a, b]$ and let P be a partition of $[a, b]$. A **Riemann sum** of f for P is any expression R_P of the form

$$R_P = \sum_{i=1}^n f(w_i) \Delta x_i$$

where w_i is some number in $[x_{i-1}, x_i]$ for $i = 1, 2, \dots, n$.

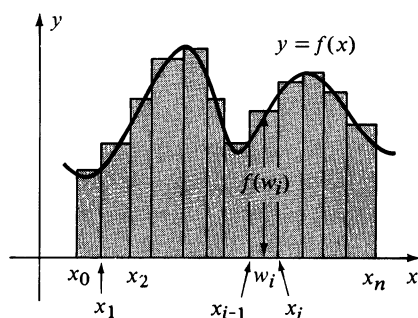


FIGURE 5.7

The sum in Definition (5.4), which represents a sum of areas of certain inscribed rectangles, is a special type of Riemann sum. In Definition (5.6), $f(w_i)$ is not necessarily a maximum or minimum value of f on $[x_{i-1}, x_i]$. Thus, if we construct a rectangle of length $f(w_i)$ and width Δx_i as illustrated in Figure 5.7, the rectangle may be neither inscribed nor circumscribed.

In general, since $f(x)$ may be negative for some x in $[a, b]$, some terms of R_P in Definition (5.6) may be negative. Consequently, a Riemann sum does not always represent a sum of areas of rectangles. It is possible to interpret a Riemann sum geometrically as follows. If R_P is defined as in (5.6), then for each subinterval $[x_{i-1}, x_i]$ let us construct a horizontal line segment through the point $(w_i, f(w_i))$, thereby obtaining a collection of rectangles. If $f(w_i)$ is positive, the rectangle lies above the x -axis as illustrated by the shaded rectangles in Figure 5.8, and the product $f(w_i) \Delta x_i$ is the area of this rectangle. If $f(w_i)$ is negative, then the rectangle lies below the x -axis as illustrated by the unshaded rectangles in Figure 5.8. In this case the product $f(w_i) \Delta x_i$ is the *negative* of the area of a rectangle. It follows that R_P is the sum of the areas of the rectangles that lie above the x -axis and the *negatives* of the areas of the rectangles that lie below the x -axis.

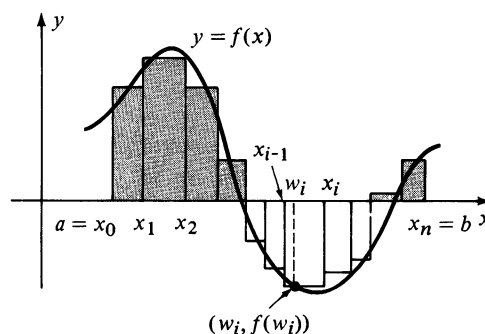


FIGURE 5.8

Example Suppose $f(x) = 8 - (x^2/2)$ and P is the partition of $[0, 6]$ into the five subintervals determined by $x_0 = 0$, $x_1 = 1.5$, $x_2 = 2.5$, $x_3 = 4.5$, $x_4 = 5$, and $x_5 = 6$. Find (a) the norm of the partition and (b) the Riemann sum R_P if $w_1 = 1$, $w_2 = 2$, $w_3 = 3.5$, $w_4 = 5$, and $w_5 = 5.5$.

Solution The graph of f is sketched in Figure 5.9. Also shown in the figure are the points on the x -axis that correspond to x_i and the rectangles of lengths $|f(w_i)|$ for $i = 1, 2, 3, 4$, and 5 .

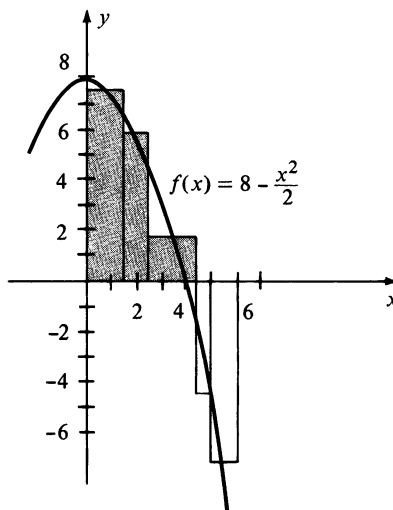


FIGURE 5.9

Thus, $\Delta x_1 = 1.5$, $\Delta x_2 = 1$, $\Delta x_3 = 2$, $\Delta x_4 = 0.5$, $\Delta x_5 = 1$

and hence the norm $\|P\|$ of the partition is Δx_3 , or 2. By Definition (5.6),

$$\begin{aligned} R_P &= f(w_1) \Delta x_1 + f(w_2) \Delta x_2 + f(w_3) \Delta x_3 + f(w_4) \Delta x_4 + f(w_5) \Delta x_5 \\ &= f(1)(1.5) + f(2)(1) + f(3.5)(2) + f(5)(0.5) + f(5.5)(1) \\ &= (7.5)(1.5) + (6)(1) + (1.875)(2) + (-4.5)(0.5) + (-7.125)(1) \end{aligned}$$

which reduces to $R_P = 11.625$. ■

In the future we shall not always specify the number n of subintervals in a partition P of $[a, b]$. In this event a Riemann sum (5.6) will be written

$$R_P = \sum_i f(w_i) \Delta x_i$$

where it is understood that terms of the form $f(w_i) \Delta x_i$ are to be summed over all subintervals $[x_{i-1}, x_i]$ of the partition P .

In a manner similar to that used in formulating Definition (5.4), we may define what is meant by

$$\lim_{\|P\| \rightarrow 0} \sum_i f(w_i) \Delta x_i = I$$

where I is a real number. Intuitively, it will mean that if the norm $\|P\|$ of the partition P is close to 0, then every Riemann sum for P is close to I .

Definition (5.7)

Let f be a function that is defined on a closed interval $[a, b]$, and let I be a real number. The statement

$$\lim_{\|P\| \rightarrow 0} \sum_i f(w_i) \Delta x_i = I$$

means that for every $\varepsilon > 0$ there exists a $\delta > 0$, such that if P is a partition of $[a, b]$ with $\|P\| < \delta$, then

$$\left| \sum_i f(w_i) \Delta x_i - I \right| < \varepsilon$$

for any choice of numbers w_i in the subintervals $[x_{i-1}, x_i]$ of P . The number I is called a **limit of a sum**.

Note that for every $\delta > 0$ there are infinitely many partitions P such that $\|P\| < \delta$. Moreover, for each such partition P there are infinitely many ways of choosing the numbers w_i in $[x_{i-1}, x_i]$. Consequently, there may be an infinite number of different Riemann sums associated with *each* partition P . However, if the limit I exists, then for any ε , each of these Riemann sums is within ε units of I , provided a small enough norm is chosen. Although Definition (5.7) differs from the definition of limit of a function, a proof similar to that given in Appendix II may be used to show that if the limit I exists, then it is unique.

We next define the definite integral as a limit of a sum.

Definition (5.8)

Let f be a function that is defined on a closed interval $[a, b]$. The **definite integral of f from a to b** , denoted by $\int_a^b f(x) dx$, is given by

$$\int_a^b f(x) dx = \lim_{\|P\| \rightarrow 0} \sum_i f(w_i) \Delta x_i$$

provided the limit exists.

If the definite integral of f from a to b exists, then f is said to be **integrable** on the closed interval $[a, b]$, or we say that the integral $\int_a^b f(x) dx$ **exists**. The process of finding the number represented by the limit is called **evaluating the integral**.

The symbol $\int$ in Definition (5.8) is called an **integral sign**. It may be thought of as an elongated letter S (the first letter of the word *sum*) and is used to indicate the connection between definite integrals and Riemann sums. The numbers a and b are referred to as the **limits of integration**, a being called the

lower limit and b the **upper limit**. Note that the word *limit* in this terminology is used in conjunction with the smallest and largest numbers in the interval $[a, b]$ and has no connection with any definitions of limits given earlier. The expression $f(x)$ which appears to the right of the integral sign (we sometimes say “*behind* the integral sign”) is called the **integrand**. Finally, the symbol dx which follows $f(x)$ should not be confused with the differential of x defined in Section 3.5. At this stage of our work it is merely used to indicate the variable. Later in the text the use of the differential symbol will have certain practical advantages.

Letters other than x may be used in the notation for the definite integral. This follows from the fact that when we describe a function, the symbol used for the independent variable is immaterial. Thus, if f is integrable on $[a, b]$, then

$$\int_a^b f(x) dx = \int_a^b f(s) ds = \int_a^b f(t) dt$$

etc. For this reason the letter x in Definition (5.8) is sometimes referred to as a **dummy variable**.

Whenever an interval $[a, b]$ is employed it is assumed that $a < b$. Consequently, Definition (5.8) does not take into account the cases in which the lower limit of integration is greater than or equal to the upper limit. The definition may be extended to include the case where the lower limit is greater than the upper limit, as follows

Definition (5.9)

$$\text{If } c > d, \text{ then } \int_c^d f(x) dx = - \int_d^c f(x) dx.$$

In words, Definition (5.9) may be phrased as “interchanging the limits of integration changes the sign of the integral.” One reason for the form of Definition (5.9) will become apparent later, after we have considered the Fundamental Theorem of Calculus.

The case in which the lower and upper limits of integration are equal is covered by the next definition.

Definition (5.10)

$$\text{If } f(a) \text{ exists, then } \int_a^a f(x) dx = 0.$$

Not every function f is integrable. For example, if $f(x)$ becomes positively or negatively infinite at some number in $[a, b]$, then the definite integral

does not exist. To illustrate, let f be defined on $[a, b]$ and $\lim_{x \rightarrow a^+} f(x) = \infty$. In the first subinterval $[x_0, x_1]$ of any partition P of $[a, b]$, a number w_1 can be found such that $f(w_1) \Delta x_1$ is larger than any given number M . It follows that for any partition P , we can form a Riemann sum $\sum_i f(w_i) \Delta x_i$ that is arbitrarily large. Hence if I is any real number and P any partition of $[a, b]$, then there exist Riemann sums R_P such that $R_P - I$ is arbitrarily large. This implies that f is not integrable. A similar argument can be given if f becomes infinite at any other number in $[a, b]$. Consequently, *if a function f is integrable on $[a, b]$, then it is bounded on $[a, b]$* ; that is, there is a real number M such that $|f(x)| \leq M$ for all x in $[a, b]$.

The reader may be tempted to conjecture that if a function is discontinuous somewhere in $[a, b]$, then it is not integrable. This conjecture is false. Definite integrals of discontinuous functions may or may not exist, depending on the nature of the discontinuities. However, according to the next theorem, continuous functions are *always* integrable.

Theorem (5.11)

If f is continuous on $[a, b]$, then f is integrable on $[a, b]$.

A proof of Theorem 5.11 may be found in texts on advanced calculus.

If f is integrable, then the limit in Definition (5.8) exists for all choices of w_i in $[x_{i-1}, x_i]$. This fact allows us to specialize w_i if we wish to do so. For example, we could always choose w_i as the smallest number x_{i-1} in the subinterval, or as the largest number x_i , or as the midpoint of the subinterval, or as the number that always produces the minimum or maximum value in $[x_{i-1}, x_i]$, etc. In addition, since the limit is independent of the partitions P of $[a, b]$ (provided that $\|P\|$ is sufficiently small) we may specialize the partitions to the case in which all the subintervals $[x_{i-1}, x_i]$ have the same length Δx . A partition of this type is called a **regular partition**. We will see in the next chapter that the specializations we have described are often used in applications. As an immediate illustration, we have the following important result.

Theorem (5.12)

If f is continuous and $f(x) \geq 0$ for all x in $[a, b]$, then the area A of the region under the graph of f from a to b is given by

$$A = \int_a^b f(x) dx.$$

Proof The area A was defined, using (5.4), as the limit of the sum $\sum_i f(u_i) \Delta x$, where $f(u_i)$ is the minimum value of f in $[x_{i-1}, x_i]$. Since this is a special type of Riemann sum, the conclusion follows from Definition (5.8). $\square$

5.2 Exercises

In Exercises 1–4, the given numbers $\{x_0, x_1, x_2, \dots, x_n\}$ determine a partition P of the indicated interval $[a, b]$. Find $\Delta x_1, \Delta x_2, \dots, \Delta x_n$ and the norm $\|P\|$ of the partition.

- 1 $[0, 5]; \{0, 1.1, 2.6, 3.7, 4.1, 5\}$
- 2 $[2, 6]; \{2, 3, 3.7, 4, 5.2, 6\}$
- 3 $[-3, 1]; \{-3, -2.7, -1, 0.4, 0.9, 1\}$
- 4 $[1, 4]; \{1, 1.6, 2, 3.5, 4\}$

In Exercises 5 and 6 find the Riemann sum R_P of f , where P is the regular partition of $[1, 5]$ into the four equal subintervals determined by $x_0 = 1, x_1 = 2, x_2 = 3, x_3 = 4, x_4 = 5$, and

- (a) w_i is the right-hand endpoint x_i of $[x_{i-1}, x_i]$.
- (b) w_i is the left-hand endpoint x_{i-1} of $[x_{i-1}, x_i]$.
- (c) w_i is the midpoint of $[x_{i-1}, x_i]$.
- 5 $f(x) = 2x + 3$
- 6 $f(x) = 3 - 4x$
- 7 If $f(x) = 8 - (x^2/2)$, find the Riemann sum R_P of f where P is the regular partition of $[0, 6]$ into the six equal subintervals determined by $x_0 = 0, x_1 = 1, x_2 = 2, x_3 = 3, x_4 = 4, x_5 = 5, x_6 = 6$, and w_i is the midpoint of the interval $[x_{i-1}, x_i]$.
- 8 If $f(x) = 8 - (x^2/2)$, find the Riemann sum R_P of f where P is the partition of $[0, 6]$ into the four subintervals determined by $x_0 = 0, x_1 = 1.5, x_2 = 3, x_3 = 4.5, x_4 = 6$, and $w_1 = 1, w_2 = 2, w_3 = 4$, and $w_4 = 5$.
- 9 Suppose $f(x) = x^3$ and P is the partition of $[-2, 4]$ into the four subintervals determined by $x_0 = -2, x_1 = 0, x_2 = 1, x_3 = 3$, and $x_4 = 4$. Find the Riemann sum R_P if $w_1 = -1, w_2 = 1, w_3 = 2$, and $w_4 = 4$.
- 10 Suppose $f(x) = \sqrt{x}$ and P is the partition of $[1, 16]$ into the five subintervals determined by $x_0 = 1, x_1 = 3, x_2 = 5, x_3 = 7, x_4 = 9$, and $x_5 = 16$. Find the Riemann sum R_P if $w_1 = 1, w_2 = 4, w_3 = 5, w_4 = 9$, and $w_5 = 9$.

In Exercises 11–14, use Definition (5.8) to express each limit as a definite integral on the indicated closed interval $[a, b]$.

- 11 $\lim_{\|P\| \rightarrow 0} \sum_{i=1}^n (3w_i^2 - 2w_i + 5) \Delta x_i; [-1, 2]$
- 12 $\lim_{\|P\| \rightarrow 0} \sum_{i=1}^n \pi(w_i^2 - 4) \Delta x_i; [2, 3]$

$$13 \quad \lim_{\|P\| \rightarrow 0} \sum_{i=1}^n 2\pi w_i (1 + w_i^3) \Delta x_i; [0, 4]$$

$$14 \quad \lim_{\|P\| \rightarrow 0} \sum_{i=1}^n (\sqrt[3]{w_i} + 4w_i) \Delta x_i; [-4, -3]$$

$$15 \quad \text{If } \int_1^4 \sqrt{x} \, dx = \frac{14}{3}, \text{ find } \int_4^1 \sqrt{x} \, dx.$$

$$16 \quad \text{Find } \int_3^3 x^2 \, dx.$$

$$17 \quad \text{If } \int_1^2 (5x^4 - 1) \, dx = 30, \text{ find } \int_1^2 (5r^4 - 1) \, dr.$$

$$18 \quad \text{If } \int_{-1}^8 \sqrt[3]{s} \, ds = \frac{45}{4}, \text{ find } \int_8^{-1} \sqrt[3]{t} \, dt.$$

In Exercises 19–22 find the value of the definite integral by interpreting it as the area under the graph of a function f .

$$19 \quad \int_{-3}^2 (2x + 6) \, dx$$

$$20 \quad \int_{-1}^2 (7 - 3x) \, dx$$

$$21 \quad \int_0^3 \sqrt{9 - x^2} \, dx$$

$$22 \quad \int_{-a}^a \sqrt{a^2 - x^2} \, dx, a > 0$$

$$23 \quad \text{Find } \int_0^5 x^3 \, dx. (\text{Hint: See Example 4 of Section 5.1.})$$

24 Let c be an arbitrary real number and suppose $f(x) = c$ for all x . If P is any partition of $[a, b]$, show that every Riemann sum R_P of f equals $c(b - a)$. Use this fact to prove that $\int_a^b c \, dx = c(b - a)$. Interpret this geometrically if $c > 0$.

25 Give an example of a function which is continuous on the interval $(0, 1)$ such that $\int_0^1 f(x) \, dx$ does not exist. Why doesn't this contradict Theorem (5.11)?

26 Give an example of a function which is not continuous on $[0, 1]$ such that $\int_0^1 f(x) \, dx$ exists.

5.3 Properties of the Definite Integral

This section contains some fundamental properties of the definite integral. Most of the proofs are rather technical and have been placed in Appendix II, where the reader may study them whenever time permits.

Theorem (5.13)

$$\int_a^b k \, dx = k(b - a)$$

Proof If f is the constant function defined by $f(x) = k$ for all x in $[a, b]$ and P is a partition of $[a, b]$, then for every Riemann sum of f ,

$$\sum_i f(w_i) \Delta x_i = \sum_i k \Delta x_i = k \sum_i \Delta x_i = k(b - a),$$

since the sum $\sum_i \Delta x_i$ is the length of the interval $[a, b]$. Consequently,

$$\left| \sum_i f(w_i) \Delta x_i - k(b - a) \right| = |k(b - a) - k(b - a)| = 0,$$

which is less than any positive number ε regardless of the size of $\|P\|$. Therefore, by Definition (5.7),

$$\lim_{\|P\| \rightarrow 0} \sum_i f(w_i) \Delta x_i = \lim_{\|P\| \rightarrow 0} \sum_i k \Delta x_i = k(b - a).$$

By Definition (5.8), this means that

$$\int_a^b f(x) \, dx = \int_a^b k \, dx = k(b - a). \quad \square$$

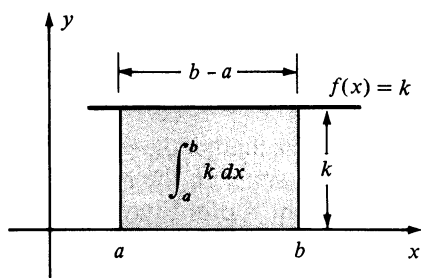


FIGURE 5.10

Theorem (5.13) is in agreement with the discussion of area in Section 5.1, for if $k > 0$, then the graph of f is a horizontal line k units above the x -axis, and the region under the graph from a to b is a rectangle with sides of length k and $b - a$ as illustrated in Figure 5.10. Hence the area $\int_a^b f(x) \, dx$ of the rectangle is $k(b - a)$.

Example 1 Evaluate $\int_{-2}^3 7 \, dx$.

Solution Using (5.13),

$$\int_{-2}^3 7 \, dx = 7[3 - (-2)] = 7(5) = 35. \quad \blacksquare$$

For the special case of (5.13) with $k = 1$ we shall abbreviate the integrand by writing

$$\int_a^b dx = b - a.$$

If a function f is integrable on $[a, b]$ and k is a real number, then by (ii) of Theorem (5.3) a Riemann sum of the function kf may be written

$$\sum_i kf(w_i) \Delta x_i = k \sum_i f(w_i) \Delta x_i.$$

It is proved in Appendix II that the limit of the sum on the left is equal to k times the limit of the sum on the right. Restating this fact in terms of definite integrals gives us the next theorem.

Theorem (5.14)

If f is integrable on $[a, b]$ and k is any real number, then kf is integrable on $[a, b]$ and

$$\int_a^b kf(x) dx = k \int_a^b f(x) dx.$$

The conclusion of Theorem (5.14) is sometimes stated “*a constant factor in the integrand may be taken outside the integral sign.*” It is *not* permissible to take expressions involving variables outside the integral sign in this manner.

If two functions f and g are defined on $[a, b]$, then by (i) of Theorem (5.3) a Riemann sum of $f + g$ may be written

$$\sum_i [f(w_i) + g(w_i)] \Delta x_i = \sum_i f(w_i) \Delta x_i + \sum_i g(w_i) \Delta x_i.$$

It can be shown that if f and g are integrable, then the limit of the sum on the left may be found by adding the limits of the two sums on the right. This fact is stated in integral form in (i) of the next theorem. A proof may be found in Appendix II. The analogous result for differences is stated in (ii).

Theorem (5.15)

If f and g are integrable on $[a, b]$, then $f + g$ and $f - g$ are integrable on $[a, b]$ and

$$(i) \quad \int_a^b [f(x) + g(x)] dx = \int_a^b f(x) dx + \int_a^b g(x) dx.$$

$$(ii) \quad \int_a^b [f(x) - g(x)] dx = \int_a^b f(x) dx - \int_a^b g(x) dx.$$

Theorem (5.15) may be extended to any finite number of functions. Thus, if $f_1, f_2, \dots, f_n$ are integrable on $[a, b]$, then so is their sum and

$$\begin{aligned} \int_a^b [f_1(x) + f_2(x) + \cdots + f_n(x)] dx \\ = \int_a^b f_1(x) dx + \int_a^b f_2(x) dx + \cdots + \int_a^b f_n(x) dx. \end{aligned}$$

Example 2 Given $\int_0^2 x^3 dx = 4$ and $\int_0^2 x dx = 2$, evaluate $\int_0^2 (5x^3 - 3x + 6) dx$.

Solution We may proceed as follows.

$$\begin{aligned} \int_0^2 (5x^3 - 3x + 6) dx &= \int_0^2 5x^3 dx - \int_0^2 3x dx + \int_0^2 6 dx \\ &= 5 \int_0^2 x^3 dx - 3 \int_0^2 x dx + 6(2 - 0) \\ &= 5(4) - 3(2) + 12 = 26 \end{aligned}$$

■

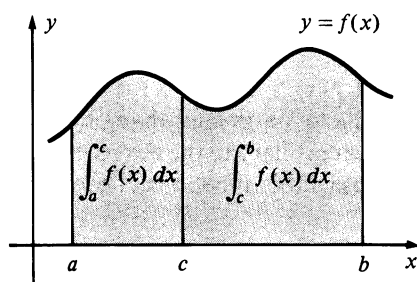


FIGURE 5.11

If f is continuous on $[a, b]$ and $f(x) \geq 0$ for all x in $[a, b]$, then by Theorem (5.12), the integral $\int_a^b f(x) dx$ is the area under the graph of f from a to b . In like manner, if $a < c < b$, then the integrals $\int_a^c f(x) dx$ and $\int_c^b f(x) dx$ are the areas under the graph of f from a to c and from c to b , respectively, as illustrated in Figure 5.11. Since the area from a to b is the sum of the two smaller areas, we have

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

The next theorem shows that the last equality is true under a more general hypothesis. The proof is given in Appendix II.

Theorem (5.16)

If $a < c < b$, and if f is integrable on both $[a, c]$ and $[c, b]$, then f is integrable on $[a, b]$ and

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

The following result shows that Theorem (5.16) can be generalized to the case where c is not necessarily between a and b .

Theorem (5.17)

If f is integrable on a closed interval and if a , b , and c are any three numbers in the interval, then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

Proof If a , b , and c are all different, then there are six possible ways of ordering these three numbers. The theorem should be verified for each of these cases as well as for the cases in which two, or all three, of the numbers are equal. We shall verify one case and leave the remaining parts as exercises. Thus, suppose the numbers are ordered such that $c < a < b$. Using Theorem (5.16),

$$\int_c^b f(x) dx = \int_c^a f(x) dx + \int_a^b f(x) dx$$

which, in turn, may be written

$$\int_a^b f(x) dx = -\int_c^a f(x) dx + \int_c^b f(x) dx.$$

The desired conclusion now follows, since interchanging the limits of integration changes the sign of the integral (see Definition (5.9)). $\square$

If f and g are continuous on $[a, b]$ and $f(x) \geq g(x) \geq 0$ for all x in $[a, b]$, then the area under the graph of f from a to b is greater than or equal to the area under the graph of g from a to b . The corollary to the next theorem is a generalization of this fact to arbitrary integrable functions. The proof of the theorem is given in Appendix II.

Theorem (5.18)

If f is integrable on $[a, b]$ and if $f(x) \geq 0$ for all x in $[a, b]$, then

$$\int_a^b f(x) dx \geq 0.$$

Corollary (5.19)

If f and g are integrable on $[a, b]$ and $f(x) \geq g(x)$ for all x in $[a, b]$, then

$$\int_a^b f(x) dx \geq \int_a^b g(x) dx.$$

Proof Since $f - g$ is integrable and $f(x) - g(x) \geq 0$ for all x in $[a, b]$, then by Theorem (5.18),

$$\int_a^b [f(x) - g(x)] dx \geq 0.$$

Applying (ii) of Theorem (5.15) leads to the desired conclusion. $\square$

It can be shown that Theorem (5.18) and Corollary (5.19) are true if $\geq$ is replaced by $>$ (see Exercise 28).

5.3 Exercises

Evaluate the definite integrals in Exercises 1–6.

1 $\int_{-2}^4 5 dx$

2 $\int_1^{10} \sqrt{2} dx$

3 $\int_6^2 3 dx$

4 $\int_4^{-3} dx$

5 $\int_{-1}^1 dx$

6 $\int_2^2 100 dx$

It will follow from our work in Section 5.5 that

$$\int_1^4 x^2 dx = 21, \quad \int_1^4 x dx = \frac{15}{2}, \quad \text{and} \quad \int_1^4 \sqrt{x} dx = \frac{14}{3}.$$

Use these facts to evaluate the integrals in Exercises 7–14.

7 $\int_1^4 (3x^2 + 5) dx$

8 $\int_1^4 (6x - 1) dx$

9 $\int_1^4 (2 - 9x - 4x^2) dx$

10 $\int_1^4 (3x + 2)^2 dx$

11 $\int_4^1 \sqrt{5x} dx$

12 $\int_1^4 2x(x + 1) dx$

13 $\int_1^4 (\sqrt{x} - 5)^2 dx$

14 $\int_4^1 (3\sqrt{x} + 1)(\sqrt{x} - 2) dx$

Verify the inequalities in Exercises 15 and 16 without evaluating the integrals.

15 $\int_1^2 (3x^2 + 4) dx \geq \int_1^2 (2x^2 + 5) dx$

16 $\int_2^4 (5x^2 - 4\sqrt{x} + 2) dx > 0$

In Exercises 17 and 18 assume that f is integrable on $[a, b]$.

17 If $f(x) \leq M$ for all x in $[a, b]$, prove that

$$\int_a^b f(x) dx \leq M(b - a).$$

Illustrate this result graphically.

18 If $m \leq f(x)$ for all x in $[a, b]$, prove that

$$m(b - a) \leq \int_a^b f(x) dx.$$

Illustrate this result graphically.

In Exercises 19–22 express each sum or difference as a single integral of the form $\int_a^b f(x) dx$.

19 $\int_5^1 f(x) dx + \int_{-3}^5 f(x) dx$

20 $\int_4^1 f(x) dx + \int_6^4 f(x) dx$

21 $\int_c^{c+h} f(x) dx - \int_c^h f(x) dx$

22 $\int_{-2}^6 f(x) dx - \int_{-2}^2 f(x) dx$

23 Prove (ii) of Theorem (5.15). (Hint: Let $f(x) - g(x) = f(x) + [-g(x)]$ and use (i) of (5.15) and Theorem (5.14).)

24 If f and g are integrable functions on $[a, b]$ and if p and q are any real numbers, prove that

$$\int_a^b [pf(x) + qg(x)] dx = p \int_a^b f(x) dx + q \int_a^b g(x) dx.$$

(Hint: Use Theorems (5.15) and (5.14).)

- 25 Complete the proof of Theorem (5.17) by considering all other orderings of the numbers a , b , and c .
- 26 Use Theorem (5.17) to prove that if f is integrable on a closed interval and if $c_1, c_2, \dots, c_n$ are any numbers in the interval, then

$$\int_{c_1}^{c_n} f(x) dx = \sum_{i=1}^{n-1} \int_{c_i}^{c_{i+1}} f(x) dx.$$

- 27 If f is integrable on $[a, b]$ and if $f(x) \leq 0$ for all x in $[a, b]$, prove that $\int_a^b f(x) dx \leq 0$. (Hint: If $f(x) \leq 0$, then $-f(x) \geq 0$.)
- 28 If f is continuous on $[a, b]$ and $f(x) > 0$ for all x in $[a, b]$, prove that $\int_a^b f(x) dx > 0$. (Hint: Let $f(u)$ be the minimum

value of f on $[a, b]$ and prove that $R_P \geq f(u)(b - a)$, where P is any partition of $[a, b]$ and R_P is any Riemann sum of f for P .)

- 29 If f is continuous on $[a, b]$, prove that

$$\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx.$$

- 30 Refer to Exercise 29. Suppose $f_1, f_2, \dots, f_n$ are continuous functions on $[a, b]$, where n is any positive integer. Use mathematical induction to prove that

$$\left| \sum_{i=1}^n \int_a^b f_i(x) dx \right| \leq \sum_{i=1}^n \int_a^b |f_i(x)| dx.$$

5.4 The Mean Value Theorem for Definite Integrals

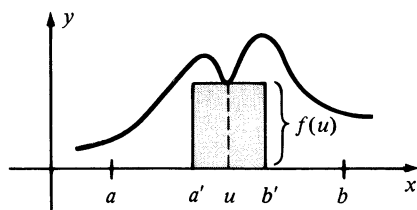


FIGURE 5.12

Suppose f is continuous and $f(x) \geq 0$ for all x in a closed interval $[a, b]$. If $f(c) > 0$ for some c in $[a, b]$, then $\lim_{x \rightarrow c} f(x) > 0$ and, by an argument similar to that used in the proof of Theorem (2.7), there is an interval $[a', b']$ contained in $[a, b]$ throughout which $f(x)$ is positive. Let $f(u)$ be the minimum value of f on $[a', b']$, as illustrated in Figure 5.12. It follows that the area under the graph of f from a' to b' is at least as large as the area $f(u)(b' - a')$ of the pictured rectangle. Consequently, by Theorem (5.12), $\int_a^b f(x) dx > 0$. This result can also be proved directly from the definition of the definite integral.

Suppose the functions f and g are continuous on $[a, b]$. If $f(x) \geq g(x)$ for all x in $[a, b]$, but $f \neq g$, then $f(x) - g(x) > 0$ for some x and, by the previous discussion, $\int_a^b [f(x) - g(x)] dx > 0$. Consequently, $\int_a^b f(x) dx > \int_a^b g(x) dx$. This fact will be used in the proof of the next theorem.

The Mean Value Theorem for Definite Integrals (5.20)

If f is continuous on a closed interval $[a, b]$, then there is a number z in the open interval (a, b) such that

$$\int_a^b f(x) dx = f(z)(b - a).$$

Proof If f is a constant function, then the result follows trivially from Theorem (5.13) where z is any number in (a, b) . Next assume that f is not a constant function and suppose that m and M are the minimum and maximum values of f , respectively, on $[a, b]$. Let $f(u) = m$ and $f(v) = M$ where u

and v are in $[a, b]$. Since f is not a constant function, $m < f(x) < M$ for some x in $[a, b]$ and hence by the remark immediately preceding this theorem,

$$\int_a^b m \, dx < \int_a^b f(x) \, dx < \int_a^b M \, dx.$$

Employing Theorem (5.13),

$$m(b-a) < \int_a^b f(x) \, dx < M(b-a).$$

Dividing by $b-a$ and replacing m and M by $f(u)$ and $f(v)$, respectively, we obtain

$$f(u) < \frac{1}{b-a} \int_a^b f(x) \, dx < f(v).$$

Since $[1/(b-a)] \int_a^b f(x) \, dx$ is a number between $f(u)$ and $f(v)$, it follows from the Intermediate Value Theorem (2.29) that there is a number z , strictly between u and v , such that

$$f(z) = \frac{1}{b-a} \int_a^b f(x) \, dx.$$

Multiplying both sides by $b-a$ gives us the conclusion of the theorem. $\square$

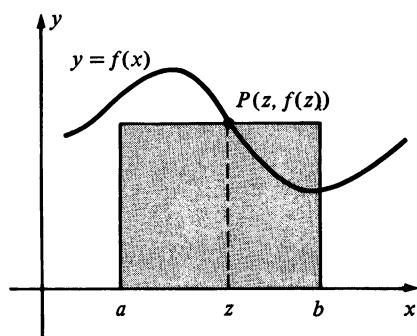


FIGURE 5.13

The number z of Theorem (5.20) is not necessarily unique. Indeed, as pointed out in the proof, if f is a constant function then *any* number z can be used. The theorem guarantees that at *least* one number z will produce the desired result.

The Mean Value Theorem has an interesting geometric interpretation if $f(x) \geq 0$ on $[a, b]$. In this case $\int_a^b f(x) \, dx$ is the area under the graph of f from a to b and the number $f(z)$ in Theorem (5.20) is the y -coordinate of the point P on the graph of f having x -coordinate z (see Figure 5.13). If a horizontal line is drawn through P , then the area of the rectangular region bounded by this line, the x -axis, and the lines $x = a$ and $x = b$ is $f(z)(b-a)$ which, according to Theorem (5.20), is the same as the area under the graph of f from a to b .

Example It can be proved that $\int_0^3 [4 - (x^2/4)] \, dx = \frac{39}{4}$. Find a number that satisfies the conclusion of the Mean Value Theorem for this integral.

Solution According to the Mean Value Theorem for Definite Integrals, there is a number z between 0 and 3 such that

$$\int_0^3 \left(4 - \frac{x^2}{4}\right) \, dx = \left(4 - \frac{z^2}{4}\right)(3 - 0)$$

$$\text{or, equivalently,} \quad \frac{39}{4} = \left(\frac{16 - z^2}{4}\right)(3).$$

Multiplying both sides of the last equation by $\frac{4}{3}$ leads to $13 = 16 - z^2$ and, therefore, $z^2 = 3$. Consequently, $\sqrt{3}$ satisfies the conclusion of Theorem (5.20). ■

The Mean Value Theorem for Definite Integrals can be used to help prove a number of important theorems. One of the most important is the *Fundamental Theorem of Calculus* given in the next section.

5.4 Exercises

The definite integrals given in Exercises 1–10 may be verified by methods to be introduced later. In each exercise find numbers that satisfy the conclusion of the Mean Value Theorem for Definite Integrals (5.20).

$$1 \quad \int_0^3 3x^2 dx = 27$$

$$2 \quad \int_{-1}^3 (3x^2 - 2x + 3) dx = 32$$

$$3 \quad \int_{-1}^8 3\sqrt{x+1} dx = 54$$

$$4 \quad \int_{-2}^{-1} 8x^{-3} dx = -3$$

$$5 \quad \int_1^2 (4x^3 - 1) dx = 14$$

$$6 \quad \int_1^4 (2 + 3\sqrt{x}) dx = 20$$

$$7 \quad \int_2^7 (x+3)^{-2} dx = 1/10$$

$$8 \quad \int_1^8 4\sqrt[3]{x} dx = 45$$

$$9 \quad \int_0^a \sqrt{a^2 - x^2} dx = \frac{\pi}{4} a^2, a > 0$$

$$10 \quad \int_1^3 (x^2 + x^{-2}) dx = 28/3$$

11 If $f(x) = k$ for all x in $[a, b]$, prove that every number z in $[a, b]$ satisfies the conclusion of Theorem (5.20). Interpret this fact geometrically.

12 If $f(x) = x$ and $0 < a < b$, find (without integrating) a number z in (a, b) such that $\int_a^b f(x) dx = f(z)(b - a)$.

5.5 The Fundamental Theorem of Calculus

The task of evaluating a definite integral by means of Definition (5.8) is quite difficult even in the simplest cases. This section contains a theorem that can be used to find the definite integral without using limits of sums. Due to its importance in evaluating definite integrals, and because it exhibits the connection between differentiation and integration, the theorem is aptly called *The Fundamental Theorem of Calculus*. This theorem was discovered independently by Sir Isaac Newton (1642–1727) in England and by Gottfried Wilhelm Leibniz (1646–1716) in Germany. It is primarily because of this discovery that these outstanding mathematicians are credited with the invention of calculus.

To avoid confusion, in the following discussion we shall use the variable t and denote the definite integral of f from a to b by $\int_a^b f(t) dt$. If f is continuous

on $[a, b]$ and if x is in $[a, b]$, then f is continuous on $[a, x]$ and hence by Theorem (5.11) f is integrable on $[a, x]$. Consequently, the equation

$$G(x) = \int_a^x f(t) dt$$

defines a function G with domain $[a, b]$, since for each x in $[a, b]$ there corresponds a unique number $G(x)$. The next theorem brings out the remarkable fact that G is an antiderivative of f . In addition, it shows how any antiderivative may be used to find a definite integral of f .

**The Fundamental Theorem
of Calculus (5.21)**

Suppose f is continuous on a closed interval $[a, b]$.

Part I If the function G is defined by

$$G(x) = \int_a^x f(t) dt$$

for all x in $[a, b]$, then G is an antiderivative of f on $[a, b]$.

Part II If F is any antiderivative of f , then

$$\int_a^b f(x) dx = F(b) - F(a).$$

Proof To establish Part I we must show that if x is in $[a, b]$, then $G'(x) = f(x)$, that is,

$$\lim_{h \rightarrow 0} \frac{G(x+h) - G(x)}{h} = f(x).$$

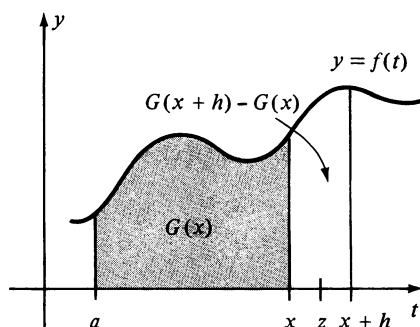


FIGURE 5.14

Before giving a formal proof, it is instructive to consider some geometric aspects of this formula. If $f(x) \geq 0$ throughout $[a, b]$, then $G(x)$ is the area under the graph of f from a to x , as illustrated in Figure 5.14. If $h > 0$, then the difference $G(x+h) - G(x)$ is the area under the graph of f from x to $x+h$, the number h is the length of the interval $[x, x+h]$, and $f(x)$ is the y -coordinate of the point with x -coordinate x on the graph of f . We will show that $[G(x+h) - G(x)]/h = f(z)$, where z is between x and $x+h$. Reasoning intuitively, it appears that if $h \rightarrow 0$, then $z \rightarrow x$ and $f(z) \rightarrow f(x)$, which is what we wish to prove.

Let us now give a rigorous proof that $G'(x) = f(x)$. If x and $x+h$ are in $[a, b]$, then using the definition of G , together with Definition (5.9) and Theorem (5.16),

$$\begin{aligned} G(x+h) - G(x) &= \int_a^{x+h} f(t) dt - \int_a^x f(t) dt \\ &= \int_a^{x+h} f(t) dt + \int_x^a f(t) dt \\ &= \int_x^{x+h} f(t) dt. \end{aligned}$$

Consequently, if $h \neq 0$,

$$\frac{G(x+h) - G(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) dt.$$

If $h > 0$, then by the Mean Value Theorem for Integrals (5.20), there is a number z (depending on h) in the open interval $(x, x+h)$ such that

$$\int_x^{x+h} f(t) dt = f(z)h$$

and, therefore,

$$\frac{G(x+h) - G(x)}{h} = f(z).$$

Since $x < z < x+h$ it follows from the continuity of f that

$$\lim_{h \rightarrow 0^+} f(z) = \lim_{z \rightarrow x^+} f(z) = f(x)$$

and hence

$$\lim_{h \rightarrow 0^+} \frac{G(x+h) - G(x)}{h} = \lim_{h \rightarrow 0^+} f(z) = f(x).$$

If $h < 0$, then we may prove in similar fashion that

$$\lim_{h \rightarrow 0^-} \frac{G(x+h) - G(x)}{h} = f(x).$$

The last two one-sided limits imply that

$$G'(x) = \lim_{h \rightarrow 0} \frac{G(x+h) - G(x)}{h} = f(x)$$

which is what we wished to prove.

To prove Part II, let F be any antiderivative of f and let G be the special antiderivative defined in Part I. It follows from Theorem (4.32) that F and G differ by a constant; that is, there is a number C such that $G(x) - F(x) = C$ for all x in $[a, b]$. Hence, from the definition of G ,

$$\int_a^x f(t) dt - F(x) = C$$

for all x in $[a, b]$. If we let $x = a$ and use the fact that $\int_a^a f(t) dt = 0$, we obtain $0 - F(a) = C$. Consequently,

$$\int_a^x f(t) dt - F(x) = -F(a).$$

Since this is an identity for all x in $[a, b]$ we may substitute b for x , obtaining

$$\int_a^b f(t) dt - F(b) = -F(a).$$

Adding $F(b)$ to both sides of this equation and replacing the variable t by x gives us the desired conclusion. $\square$

It is customary to denote the difference $F(b) - F(a)$ either by the symbol $F(x)]_a^b$ or by $[F(x)]_a^b$. Part II of the Fundamental Theorem may then be expressed as follows.

Theorem (5.22)

If f is continuous on $[a, b]$ and F is any antiderivative of f , then

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a).$$

The formula in Theorem (5.22) is also valid if $a \geq b$. Thus, if $a > b$, then by Definition (5.9),

$$\begin{aligned} \int_a^b f(x) dx &= - \int_b^a f(x) dx \\ &= -[F(a) - F(b)] \\ &= F(b) - F(a). \end{aligned}$$

If $a = b$, then by Definition (5.10),

$$\int_a^a f(x) dx = 0 = F(a) - F(a).$$

Example 1 Evaluate $\int_{-2}^3 (6x^2 - 5) dx$.

Solution An antiderivative of $6x^2 - 5$ is given by $F(x) = 2x^3 - 5x$. Applying Theorem (5.22),

$$\begin{aligned} \int_{-2}^3 (6x^2 - 5) dx &= \left[2x^3 - 5x \right]_{-2}^3 \\ &= [2(3)^3 - 5(3)] - [2(-2)^3 - 5(-2)] \\ &= [54 - 15] - [-16 + 10] = 45. \end{aligned} \quad \blacksquare$$

Note that if $F(x) + C$ is used in place of $F(x)$ in Theorem (5.22), the same result is obtained, since

$$\begin{aligned} \left[F(x) + C \right]_a^b &= \{F(b) + C\} - \{F(a) + C\} \\ &= F(b) - F(a) = \left[F(x) \right]_a^b. \end{aligned}$$

This is in keeping with the statement that *any* antiderivative F may be employed in the Fundamental Theorem. Also note that for any number k ,

$$\left[kF(x) \right]_a^b = kF(b) - kF(a) = k[F(b) - F(a)] = k \left[F(x) \right]_a^b;$$

that is, a constant factor can be “taken out” of the bracket when using this notation. This result is analogous to Theorem (5.14) on definite integrals.

Theorem (5.23)

If k is a real number, and r is a rational number such that $r \neq -1$, then

$$\int_a^b kx^r dx = \left[\left(\frac{k}{r+1} \right) x^{r+1} \right]_a^b = \left(\frac{k}{r+1} \right) (b^{r+1} - a^{r+1}).$$

Proof If $f(x) = kx^r$, then the function F defined by $F(x) = [k/(r+1)]x^{r+1}$ is an antiderivative of f , whenever f is defined. Applying Theorem (5.22) gives us the conclusion. $\square$

If an integrand is a sum of terms of the form kx^r where $r \neq -1$, then (5.23) may be applied to each term, as illustrated in the next example.

Example 2 Evaluate $\int_{-1}^2 (x^3 + 1)^2 dx$.

Solution Squaring the integrand and then applying Theorem (5.23) to each term gives us

$$\begin{aligned} \int_{-1}^2 (x^3 + 1)^2 dx &= \int_{-1}^2 (x^6 + 2x^3 + 1) dx \\ &= \left[\frac{1}{7}x^7 + \frac{2}{4}x^4 + x \right]_{-1}^2 \\ &= \left[\frac{1}{7}(2)^7 + \frac{1}{2}(2)^4 + 2 \right] - \left[\frac{1}{7}(-1)^7 + \frac{1}{2}(-1)^4 + (-1) \right] \\ &= \frac{405}{14}. \quad \blacksquare \end{aligned}$$

In the preceding example it is very important to note that

$$\int_{-1}^2 (x^3 + 1)^2 dx \neq \frac{(x^3 + 1)^3}{3} \Big|_{-1}^2.$$

Example 3 Evaluate $\int_1^4 \left(5x - 2\sqrt{x} + \frac{32}{x^3} \right) dx$.

Solution We begin by changing the form of the integrand so that Theorem (5.23) may be applied to each term. Thus

$$\begin{aligned} \int_1^4 (5x - 2x^{1/2} + 32x^{-3}) dx &= \left[\frac{5}{2}x^2 - \frac{2}{(3/2)}x^{3/2} + \frac{32}{-2}x^{-2} \right]_1^4 \\ &= \left[\frac{5}{2}x^2 - \frac{4}{3}x^{3/2} - \frac{16}{x^2} \right]_1^4 \\ &= \left[\frac{5}{2}(4)^2 - \frac{4}{3}(4)^{3/2} - \frac{16}{4^2} \right] - \left[\frac{5}{2} - \frac{4}{3} - 16 \right] \\ &= 259/6. \quad \blacksquare \end{aligned}$$

Example 4 Evaluate $\int_{-2}^3 |x| dx$.

Solution By Definition (1.2), $|x| = -x$ if $x < 0$ and $|x| = x$ if $x \geq 0$. This suggests that we use Theorem (5.16) to express the given integral as a sum of two definite integrals as follows:

$$\begin{aligned} \int_{-2}^3 |x| dx &= \int_{-2}^0 |x| dx + \int_0^3 |x| dx \\ &= \int_{-2}^0 (-x) dx + \int_0^3 x dx \\ &= -\left[\frac{x^2}{2} \right]_{-2}^0 + \left[\frac{x^2}{2} \right]_0^3 \\ &= -\left[0 - \frac{4}{2} \right] + \left[\frac{9}{2} - 0 \right] \\ &= 2 + \frac{9}{2} = \frac{13}{2}. \quad \blacksquare \end{aligned}$$

Using the antiderivative formulas given in Theorem (4.36) together with Part II of the Fundamental Theorem (5.21) leads to the following definite integral formulas for the sine and cosine functions.

Theorem (5.24)

$$\begin{aligned} \int_a^b \sin kx dx &= -\frac{1}{k} \cos kx \Big|_a^b \\ \int_a^b \cos kx dx &= \frac{1}{k} \sin kx \Big|_a^b \end{aligned}$$

Example 5 Find the area A under the graph of $y = \sin 2x$ from $x = 0$ to $x = \pi/4$.

Solution By Theorems (5.12) and (5.24),

$$\begin{aligned} A &= \int_0^{\pi/4} \sin 2x \, dx = -\frac{1}{2} \cos 2x \Big|_0^{\pi/4} \\ &= -\frac{1}{2} \left[\cos 2\left(\frac{\pi}{4}\right) - \cos 2(0) \right] \\ &= -\frac{1}{2} \left[\cos \frac{\pi}{2} - \cos 0 \right] = -\frac{1}{2} [0 - 1] = \frac{1}{2}. \quad \blacksquare \end{aligned}$$

The technique of defining a function by means of a definite integral, as in Part I of the Fundamental Theorem of Calculus (5.21), will have a very important application in Chapter 7 (see Definition (7.3)). Recall, from (5.21), that if f is continuous on $[a, b]$ and the function G is defined by $G(x) = \int_a^x f(t) \, dt$, where $a \leq x \leq b$, then G is an antiderivative of f ; that is, $D_x G(x) = f(x)$. This may be stated in integral form as follows:

$$D_x \int_a^x f(t) \, dt = f(x).$$

The preceding formula is generalized in the next theorem.

Theorem (5.25)

Let f be continuous on $[a, b]$. If $a \leq c \leq b$, then

$$D_x \int_c^x f(t) \, dt = f(x)$$

for all x in $[a, b]$.

Proof If F is an antiderivative of f , then

$$\begin{aligned} D_x \int_c^x f(t) \, dt &= D_x [F(x) - F(c)] \\ &= D_x F(x) - D_x F(c) \\ &= f(x) - 0 = f(x). \quad \square \end{aligned}$$

Example 6 If $G(x) = \int_1^x \frac{1}{t} \, dt$, where $x > 0$, find $G'(x)$.

Solution Let $f(x) = 1/x$ and consider any interval $[a, b]$ where $a > 0$ and $b \geq 1$. If $a \leq x \leq b$, then by Theorem (5.25),

$$G'(x) = D_x \int_1^x \frac{1}{t} \, dt = \frac{1}{x}. \quad \blacksquare$$

5.5 Exercises

Evaluate the definite integrals in Exercises 1–32.

$$1 \quad \int_1^4 (x^2 - 4x - 3) dx$$

$$2 \quad \int_{-2}^3 (5 + x - 6x^2) dx$$

$$3 \quad \int_{-2}^3 (8z^3 + 3z - 1) dz$$

$$4 \quad \int_0^2 (w^4 - 2w^3) dw$$

$$5 \quad \int_7^{12} dx$$

$$6 \quad \int_{-6}^{-1} 8 dx$$

$$7 \quad \int_1^2 [5/(8x^6)] dx$$

$$8 \quad \int_1^4 \sqrt{16x^3} dx$$

$$9 \quad \int_4^9 \frac{t-3}{\sqrt{t}} dt$$

$$10 \quad \int_{-1}^{-2} \frac{2s-7}{s^3} ds$$

$$11 \quad \int_{-8}^8 (\sqrt[3]{s^2} + 2) ds$$

$$12 \quad \int_1^0 t^2(\sqrt[3]{t} - \sqrt{t}) dt$$

$$13 \quad \int_0^1 (2x-3)(5x+1) dx$$

$$14 \quad \int_{-1}^1 (x^2 + 1)^2 dx$$

$$15 \quad \int_{-1}^0 (2w+3)^2 dw$$

$$16 \quad \int_8^5 \sqrt[3]{x^2 + \sqrt{x^5 + 1}} dx$$

$$17 \quad \int_3^2 \frac{x^2 - 1}{x - 1} dx$$

$$18 \quad \int_0^{-1} \frac{x^3 + 8}{x + 2} dx$$

$$19 \quad \int_1^1 (4x^2 - 5)^{100} dx$$

$$20 \quad \int_1^2 (4u^{-3} + 6u^{-4}) du$$

$$21 \quad \int_1^3 \frac{2x^3 - 4x^2 + 5}{x^2} dx$$

$$22 \quad \int_{-2}^{-1} \left(r - \frac{1}{r}\right)^2 dr$$

$$23 \quad \int_{-3}^6 |x - 4| dx$$

$$24 \quad \int_{-1}^1 (x+1)(x+2)(x+3) dx$$

$$25 \quad \int_0^4 \sqrt{3t}(\sqrt{t} + \sqrt{3}) dt$$

$$26 \quad \int_{-1}^5 |2x - 3| dx$$

$$27 \quad \int_{\pi/2}^{\pi} \cos(x/3) dx$$

$$28 \quad \int_0^{\pi/2} 3 \sin(x/2) dx$$

$$29 \quad \int_{\pi/4}^{\pi/3} (4 \sin 2\theta + 6 \cos 3\theta) d\theta$$

$$30 \quad \int_{\pi/6}^{\pi/4} (1 - \cos 4t) dt$$

$$31 \quad \int_{-\pi/6}^{\pi/6} (x + \sin 5x) dx$$

$$32 \quad \int_{-\pi/4}^0 (\sin x + \cos x)^2 dx$$

Verify the identities in Exercises 33 and 34 by first using the Fundamental Theorem of Calculus (5.21) and then differentiating.

$$33 \quad D_x \int_0^x (t^3 - 4\sqrt{t} + 5) dt = x^3 - 4\sqrt{x} + 5 \text{ if } x \geq 0$$

$$34 \quad D_x \int_0^x (5t + 3)^2 dt = (5x + 3)^2$$

$$35 \quad \text{Find } D_x \int_0^x \frac{1}{t+1} dt$$

36 Find $D_x \int_0^x \frac{1}{\sqrt{1-t^2}} dt$, if $|x| < 1$.

37 If $f(x) = x^2 + 1$, find the area of the region under the graph of f from -1 to 2 .

38 If $f(x) = x^3$, find the area of the region under the graph of f from 1 to 3 .

Verify the formulas in Exercises 39–42.

39 $\int_a^b \frac{x}{\sqrt{(x^2+1)^3}} dx = \frac{-1}{\sqrt{x^2+1}} \Big|_a^b$

40 $\int_a^b \frac{x^3}{\sqrt{x^2+1}} dx = \left[\frac{1}{3} \sqrt{(x^2+1)^3} - \sqrt{x^2+1} \right]_a^b$

41 $\int_a^b \frac{x}{(x^2+c^2)^{n+1}} dx = \frac{-1}{2n(x^2+c^2)^n} \Big|_a^b, \quad n \neq 0$

42 $\int_a^b \frac{x^2}{(x^3+c^3)^{n+1}} dx = \frac{-1}{3n(x^3+c^3)^n} \Big|_a^b, \quad n \neq 0$

Find numbers that satisfy the conclusion of the Mean Value Theorem for Integrals (5.20) for the definite integrals in Exercises 43–46.

43 $\int_0^4 (\sqrt{x} + 1) dx$

44 $\int_{-1}^1 (2x + 1)^2 dx$

45 $\int_{-1}^2 (3x^3 + 2) dx$

46 $\int_1^9 (3/x^2) dx$

47 If f is continuous on $[a, b]$, then the **average value of f on $[a, b]$** is defined as the number

$$\frac{1}{b-a} \int_a^b f(x) dx.$$

Suppose a point P moves on a coordinate line with a continuous velocity function v . Show that the average velocity during the time interval $[a, b]$ equals the average value of v on $[a, b]$.

48 Refer to Exercise 47. If a function f has a continuous derivative on $[a, b]$, show that the average rate of change of $f(x)$ with respect to x on $[a, b]$ (see Section 4.7) equals the average value of f' on $[a, b]$.

In Exercises 49 and 50, find the average value of the function f on the given interval.

49 $f(x) = x^2 + 3x - 1, [-1, 2]$

50 $f(x) = \sin x, [0, \pi]$

51 If g is differentiable and f is continuous for all x , prove that $D_x \int_a^{g(x)} f(t) dt = f(g(x))g'(x)$. (Hint: Use Part I of Theorem (5.21) and the Chain Rule.)

52 Use Exercise 51 to find $D_x \int_0^{x^2} \sqrt[3]{t^4 + 1} dt$.

5.6 Indefinite Integrals and Change of Variables

The connection between antiderivatives and definite integrals provided by the Fundamental Theorem of Calculus has made it customary to use integral signs to denote antiderivatives. In order to distinguish antiderivatives from definite integrals, no limits of integration are attached to the integral sign, as in the following definition.

Definition (5.26)

The **indefinite integral** $\int f(x) dx$ of f (or of $f(x)$) is given by

$$\int f(x) dx = F(x) + C$$

where $F'(x) = f(x)$ and C is an arbitrary constant.

Note that the indefinite integral is merely another way of specifying the most general antiderivative of f . Instead of using the terminology *anti-differentiation* for the process of finding F when f is given, we now use the phrase **indefinite integration**. The arbitrary constant C is called the **constant of integration**, $f(x)$ is called the **integrand**, and x is called the **variable of integration**. We often refer to the process of finding $F(x) + C$ in Definition (5.26) as **evaluating the indefinite integral**. The domain of F will not usually be stated explicitly. It is always assumed that a suitable interval over which f is integrable has been chosen. In particular, a closed interval on which f is continuous could be used. As with definite integrals, the symbol employed for the variable of integration is insignificant since, for example, $\int f(t) dt$, $\int f(u) du$, etc., give rise to the same function F as $\int f(x) dx$.

Since the indefinite integral of f is an antiderivative, the Fundamental Theorem of Calculus gives us the following relationship between definite and indefinite integrals.

Theorem (5.27)

$$\int_a^b f(x) dx = \left[\int f(x) dx \right]_a^b$$

Thus, if the indefinite integral of a function f is known, then definite integrals of f can be evaluated. In future chapters, methods for finding indefinite integrals of various functions will be developed. At the present time it is possible to state several useful rules. The following **power rule for (indefinite) integration** may be proved by differentiating the expression on the right-hand side of the equation and showing that the integrand is obtained.

Theorem (5.28)

$$\int x^r dx = \left(\frac{1}{r+1} \right) x^{r+1} + C$$

where the exponent r is a rational number and $r \neq -1$.

Example 1 Evaluate the following:

$$(a) \int x^2 dx \quad (b) \int \left(8t^3 - 5 + \frac{1}{t^3} \right) dt$$

Solution We need only find an antiderivative for each integrand and then add the arbitrary constant C . Thus

$$(a) \int x^2 dx = \frac{1}{3}x^3 + C.$$

$$(b) \int \left(8t^3 - 5 + \frac{1}{t^2} \right) dt = 2t^4 - 5t - \frac{1}{t} + C. \quad \blacksquare$$

The next theorem indicates what happens if an indefinite integration is followed by a differentiation, or vice versa.

Theorem (5.29)

$$\begin{aligned} \text{(i)} \quad D_x \int f(x) dx &= f(x) \\ \text{(ii)} \quad \int D_x[f(x)] dx &= f(x) + C \end{aligned}$$

Proof To prove (i) we may use Theorem (5.26) as follows:

$$D_x \int f(x) dx = D_x[F(x) + C] = F'(x) = f(x).$$

The formula in (ii) follows from the fact that $f(x) + C$ is an antiderivative of $D_x[f(x)]$. $\square$

The following theorem is analogous to Theorems (5.14) and (5.15) for definite integrals.

Theorem (5.30)

$$\begin{aligned} \text{(i)} \quad \int kf(x) dx &= k \int f(x) dx \\ \text{(ii)} \quad \int [f(x) + g(x)] dx &= \int f(x) dx + \int g(x) dx \end{aligned}$$

Proof To prove (i) we merely differentiate the right-hand side and show that the integrand on the left is obtained. Thus,

$$D_x \left[k \int f(x) dx \right] = k \cdot D_x \int f(x) dx = kf(x)$$

where we have used (i) of (5.29). Part (ii) may be proved in similar fashion. $\square$

Every rule for differentiation can be transformed into a corresponding rule for indefinite integration. For example,

$$D_x \sqrt{x^2 + 5} = \frac{x}{\sqrt{x^2 + 5}} \quad \text{implies that} \quad \int \frac{x}{\sqrt{x^2 + 5}} dx = \sqrt{x^2 + 5} + C;$$

$$D_x(x^3 + 2x)^{10} = 10(x^3 + 2x)^9(3x^2 + 2) \quad \text{implies that}$$

$$\int 10(x^3 + 2x)^9(3x^2 + 2) dx = (x^3 + 2x)^{10} + C.$$

A similar technique may be used in conjunction with the Chain Rule to obtain a useful formula for indefinite integration. Thus, suppose F is an antiderivative of a function f , and g is a differentiable function such that $g(x)$ is in the domain of F for every x in some closed interval $[a, b]$. We may then consider the composite function defined by $F(g(x))$ for all x in $[a, b]$. Applying the Chain Rule (3.30) and the fact that $F' = f$, we obtain

$$D_x F(g(x)) = F'(g(x))g'(x) = f(g(x))g'(x).$$

This, in turn, gives us the integration formula

$$\int f(g(x))g'(x) dx = F(g(x)) + C, \quad \text{where } F' = f.$$

There is a simple way to remember this formula. If we let $u = g(x)$ and formally replace $g'(x) dx$ by the differential du , we obtain

$$\int f(u) du = F(u) + C, \quad \text{where } F' = f$$

which has the same form as Theorem (5.26). This memorization device indicates that $g'(x) dx$ may be regarded as the product of $g'(x)$ and dx , and is one of the main reasons for using the symbol dx in the integral notation. Finding indefinite integrals in this way will be referred to as a **change of variable** or as **the method of substitution**. We may summarize our discussion as follows, where f and g are assumed to have the properties described previously.

Theorem (5.31)

Given $\int f(g(x))g'(x) dx$, let $u = g(x)$ and $du = g'(x) dx$. If F is an antiderivative of f , then

$$\int f(g(x))g'(x) dx = \int f(u) du = F(u) + C = F(g(x)) + C.$$

As a first application, we may extend Theorem (5.28) to powers of functions. Thus, if $f(x) = x^r$ in (5.31), then $f(g(x)) = [g(x)]^r$. If we let

$$F(x) = \frac{1}{r+1} x^{r+1}, \quad \text{then} \quad F(g(x)) = \frac{1}{r+1} [g(x)]^{r+1}$$

and the conclusion of Theorem (5.31) takes on the form

$$\int [g(x)]^r g'(x) dx = \frac{1}{r+1} [g(x)]^{r+1} + C.$$

This may be checked by differentiating the expression on the right-hand side of the equation. A convenient way of expressing the last formula is stated in the following theorem.

Theorem (5.32)

Suppose $u = g(x)$, where g is a differentiable function. If r is a rational number such that $r \neq -1$, then

$$\int u^r du = \frac{1}{r+1} u^{r+1} + C.$$

Example 2 Find $\int (2x^3 + 1)^7 x^2 dx$.

Solution If we let $u = 2x^3 + 1$, then $du = 6x^2 dx$. To obtain the form (5.32), it is necessary to introduce the factor 6 in the integrand. Doing this, and compensating by multiplying the integral by $1/6$ (this is legitimate by (i) of Theorem (5.30)), we obtain

$$\int (2x^3 + 1)^7 x^2 dx = \frac{1}{6} \int (2x^3 + 1)^7 6x^2 dx.$$

Making the indicated substitution and integrating,

$$\int (2x^3 + 1)^7 x^2 dx = \frac{1}{6} \int u^7 du = \frac{1}{6} \left[\frac{1}{8} u^8 + K \right]$$

where K is a constant. It is now necessary to return to the original variable x . Since $u = 2x^3 + 1$, the last equality gives us

$$\begin{aligned} \int (2x^3 + 1)^7 x^2 dx &= \frac{1}{6} \left[\frac{1}{8} (2x^3 + 1)^8 + K \right] \\ &= \frac{1}{48} (2x^3 + 1)^8 + \frac{1}{6} K. \end{aligned}$$

Instead of employing constants of integration such as $\frac{1}{6}K$ it is customary to write this result as

$$\int (2x^3 + 1)^7 x^2 dx = \frac{1}{48} (2x^3 + 1)^8 + C.$$

The relationship between K and C is $C = \frac{1}{6}K$; however, there is no practical advantage in remembering this fact. In the future we shall manipulate constants of integration in various ways without making explicit mention of relationships that exist. Moreover, instead of proceeding as above, it should be clear that it is permissible to integrate as follows:

$$\frac{1}{6} \int u^7 du = \frac{1}{6} \left[\frac{1}{8} u^8 \right] + C.$$

Formal substitutions in indefinite integrals can be made in different ways. To illustrate, another method for solving Example 2 is to let

$$u = 2x^3 + 1, \quad du = 6x^2 dx, \quad \frac{1}{6} du = x^2 dx.$$

We then substitute directly for $x^2 dx$ instead of introducing the number 6 in the integrand as follows:

$$\begin{aligned}\int (2x^3 + 1)^7 x^2 dx &= \int u^7 \frac{1}{6} du = \frac{1}{6} \int u^7 du \\ &= \frac{1}{48} u^8 + C = \frac{1}{48} (2x^3 + 1)^8 + C. \quad \blacksquare\end{aligned}$$

The change of variable technique for indefinite integrals is a powerful tool, provided the student can recognize that the integrand is of the form $f(g(x))g'(x)$, or $f(g(x))kg'(x)$ where k is a real number. The ability to recognize this form is directly proportional to the number of exercises worked!

Example 3 Find $\int x\sqrt[3]{7 - 6x^2} dx$.

Solution Note that the integrand contains the term $x dx$. If the factor x were missing, the change of variable technique could not be applied. Letting

$$u = 7 - 6x^2, \quad du = -12x dx,$$

introducing the factor -12 in the integrand, and then compensating by multiplying the integral by $-\frac{1}{12}$, gives us

$$\begin{aligned}\int x\sqrt[3]{7 - 6x^2} dx &= -\frac{1}{12} \int \sqrt[3]{7 - 6x^2} (-12)x dx \\ &= -\frac{1}{12} \int \sqrt[3]{u} du = -\frac{1}{12} \int u^{1/3} du \\ &= -\frac{1}{12} \frac{u^{4/3}}{(4/3)} + C = -\frac{1}{16} (7 - 6x^2)^{4/3} + C.\end{aligned}$$

As in the second solution to Example 2, we could begin with $du = -12x dx$ and write $x dx = (-\frac{1}{12}) du$. The change of variables then takes on the form

$$\int \sqrt[3]{7 - 6x^2} x dx = \int \sqrt[3]{u} \left(-\frac{1}{12}\right) du = -\frac{1}{12} \int \sqrt[3]{u} du.$$

The remainder of the solution now proceeds exactly as before. $\blacksquare$

Letting f in Theorem (5.31) denote either the sine or cosine functions gives us the following theorem.

Theorem (5.33)

If $u = g(x)$, where g is differentiable, then

$$\int \sin u du = -\cos u + C$$

$$\int \cos u du = \sin u + C.$$

The preceding theorem generalizes the antiderivative formulas for $\sin kx$ and $\cos kx$ given in Theorems (4.36) and (5.24). Indeed, those formulas may now be discarded in favor of the technique used in the next example.

Example 4 Evaluate $\int \sin 5x \, dx$.

Solution We let $u = 5x$ so that $du = 5 \, dx$, and proceed as follows.

$$\begin{aligned}\int \sin 5x \, dx &= \frac{1}{5} \int (\sin 5x) 5 \, dx \\ &= \frac{1}{5} \int \sin u \, du \\ &= -\frac{1}{5} \cos u + C \\ &= -\frac{1}{5} \cos 5x + C\end{aligned}$$

■

Example 5 Evaluate $\int \frac{\cos \sqrt{x}}{\sqrt{x}} \, dx$.

Solution If we let $u = \sqrt{x} = x^{1/2}$, then

$$du = \frac{1}{2} x^{-1/2} \, dx = \frac{1}{2\sqrt{x}} \, dx,$$

and the integral may be evaluated as follows:

$$\begin{aligned}\int \frac{\cos \sqrt{x}}{\sqrt{x}} \, dx &= 2 \int \cos \sqrt{x} \left(\frac{1}{2\sqrt{x}} \right) \, dx \\ &= 2 \int \cos u \, du = 2 \sin u + C \\ &= 2 \sin \sqrt{x} + C.\end{aligned}$$

■

The method of substitution may also be used to evaluate definite integrals. We could use Theorem (5.31) to find an indefinite integral (that is, an antiderivative) and then apply the Fundamental Theorem of Calculus. Another method, which is sometimes shorter, is to change the limits of integration. Using (5.31) together with the Fundamental Theorem gives us the following formula, where $F' = f$:

$$\int_a^b f(g(x))g'(x) \, dx = F(g(x)) \Big|_a^b.$$

The number on the right of this equality may be written

$$F(g(b)) - F(g(a)) = F(u) \Big|_{g(a)}^{g(b)} = \int_{g(a)}^{g(b)} f(u) du.$$

This gives us the following important result, where for brevity we have not restated the restrictions on f and g .

Theorem (5.34)

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du, \quad \text{where } u = g(x).$$

Theorem (5.34) states that after making the substitutions $u = g(x)$ and $du = g'(x) dx$, we may use the values of g which correspond to $x = a$ and $x = b$, respectively, as the limits of the integral involving u . It is then unnecessary to return to the variable x after integrating. This technique is illustrated in the next example.

Example 6 Evaluate $\int_2^{10} \frac{3}{\sqrt{5x-1}} dx$.

Solution Let us begin by writing the integral as

$$3 \int_2^{10} \frac{1}{\sqrt{5x-1}} dx.$$

The form of the integrand suggests the substitution $u = 5x - 1$. Of course,

$$\text{if } u = 5x - 1, \text{ then } du = 5 dx.$$

To change the limits of integration we note that

$$\text{if } x = 2, \text{ then } u = 9,$$

$$\text{and if } x = 10, \text{ then } u = 49.$$

Applying Theorem (5.34) leads to the following evaluation:

$$\begin{aligned} 3 \int_2^{10} \frac{1}{\sqrt{5x-1}} dx &= \frac{3}{5} \int_9^{49} \frac{1}{\sqrt{5x-1}} 5 dx \\ &= \frac{3}{5} \int_9^{49} \frac{1}{\sqrt{u}} du = \frac{3}{5} \int_9^{49} u^{-1/2} du \\ &= \left(\frac{3}{5} \right) 2u^{1/2} \Big|_9^{49} = \frac{6}{5} [49^{1/2} - 9^{1/2}] \\ &= 24/5. \end{aligned}$$

■

5.6 Exercises

Evaluate the integrals in Exercises 1–33.

1 $\int (3x + 1)^4 dx$

2 $\int (2x^2 - 3)^5 x dx$

3 $\int \sqrt{t^3 - 1} t^2 dt$

4 $\int \sqrt{9 - z^2} z dz$

5 $\int \frac{x - 2}{(x^2 - 4x + 3)^3} dx$

6 $\int \frac{x^2 + x}{(4 - 3x^2 - 2x^3)^4} dx$

7 $\int \frac{s}{\sqrt[3]{1 - 2s^2}} ds$

8 $\int \sqrt[5]{t^4 - t^2} (10t^3 - 5t) dt$

9 $\int \frac{(\sqrt{u} + 3)^4}{\sqrt{u}} du$

10 $\int \left(1 + \frac{1}{u}\right)^{-3} \left(\frac{1}{u^2}\right) du$

11 $\int_1^4 \sqrt{5 - x} dx$

12 $\int_1^5 \sqrt[3]{2x - 1} dx$

13 $\int \cos(4x - 3) dx$

14 $\int \sin(1 + 6x) dx$

15 $\int x \sin(x^2) dx$

16 $\int \frac{\cos \sqrt[3]{x}}{\sqrt[3]{x^2}} dx$

17 $\int \cos 3x \sqrt[3]{\sin 3x} dx$

18 $\int \frac{\sin 2x}{\sqrt{1 - \cos 2x}} dx$

19 $\int_{-1}^1 (t^2 - 1)^3 t dt$

20 $\int_{-2}^0 \frac{v^2}{(v^3 - 2)^2} dv$

21 $\int_0^1 \frac{1}{(3 - 2v)^2} dv$

22 $\int_0^4 \frac{x}{\sqrt{x^2 + 9}} dx$

23 $\int (x^2 + 1)^3 dx$

24 $\int (3 - x^3)^2 x dx$

25 $\int 5\sqrt{8x + 5} dx$

26 $\int \frac{6}{\sqrt{4 - 5t}} dt$

27 $\int_1^4 \frac{1}{\sqrt{x}(\sqrt{x} + 1)^3} dx$

28 $\int (3 - x^4)^3 x^3 dx$

29 $\int \sin x (1 + \sqrt{\cos x})^2 dx$

30 $\int \sin^3 x \cos x dx$

31 $\int \frac{\sin x}{\cos^2 x} dx$

32 $\int \frac{\cos x}{(1 - \sin x)^2} dx$

33 $\int (2 + 5 \cos x)^3 \sin x dx$

34 Show, by evaluating in three different ways, that

$$\begin{aligned} \int \sin x \cos x dx &= \frac{1}{2} \sin^2 x + C \\ &= -\frac{1}{2} \cos^2 x + D = -\frac{1}{4} \cos 2x + E. \end{aligned}$$

How can the answers be reconciled?

Evaluate the integrals in Exercises 35–38 by (a) the method of substitution and (b) expanding the integrand. In what way do the constants of integration differ?

$$35 \int (x+4)^2 dx \qquad 36 \int (x^2+4)^2 x dx$$

$$37 \int \frac{(\sqrt{x}+3)^2}{\sqrt{x}} dx \qquad 38 \int \left(1 + \frac{1}{x}\right)^2 \frac{1}{x^2} dx$$

Verify the identities in Exercises 39 and 40 by first integrating and then differentiating.

$$39 \quad D_x \int x^3 \sqrt{x^4+5} dx = x^3 \sqrt{x^4+5}$$

$$40 \quad D_x \int (3x+2)^7 dx = (3x+2)^7$$

$$41 \quad \text{Find } D_x \int \frac{1}{\sqrt{x^3+x+5}} dx.$$

$$42 \quad \text{Find } \int D_x \frac{x}{\sqrt[3]{x+1}} dx.$$

$$43 \quad \text{Find } \int_0^3 D_x \sqrt{x^2+16} dx.$$

$$44 \quad \text{Find } D_x \int_0^1 \sqrt{x^2+4x} dx.$$

$$45 \quad \text{If } f(x) = \sqrt{x+1}, \text{ find the area of the region under the graph of } f \text{ from 0 to 3.}$$

$$46 \quad \text{If } f(x) = x/(x^2+1)^2, \text{ find the area of the region under the graph of } f \text{ from 1 to 2.}$$

In Exercises 47–50 find numbers that satisfy the conclusion of the Mean Value Theorem for Integrals.

$$47 \quad \int_0^4 \frac{x}{\sqrt{x^2+9}} dx \qquad 48 \quad \int_{-2}^0 \sqrt[3]{x+1} dx$$

$$49 \quad \int_0^5 \sqrt{x+4} dx \qquad 50 \quad \int_{-3}^2 \sqrt{6-x} dx$$

Verify the formulas in Exercises 51–54.

$$51 \quad \int \frac{\sqrt{x^2-a^2}}{x^4} dx = \frac{\sqrt{(x^2-a^2)^3}}{3a^2x^3} + C$$

$$52 \quad \int \frac{1}{x^2\sqrt{x^2-a^2}} dx = \frac{\sqrt{x^2-a^2}}{a^2x} + C$$

$$53 \quad \int \sin^2 x dx = \frac{1}{2}x - \frac{1}{4}\sin 2x + C$$

$$54 \quad \int \cos^3 x dx = \frac{1}{3}(2 + \cos^2 x) \sin x + C$$

$$55 \quad \text{Let } f \text{ be continuous on } [-a, a]. \text{ If } f \text{ is an even function, show that } \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx \text{ and interpret this result geometrically. Verify the result for the special case } f(x) = \cos x.$$

$$56 \quad \text{Let } f \text{ be continuous on } [-a, a]. \text{ If } f \text{ is an odd function show that } \int_{-a}^a f(x) dx = 0 \text{ and interpret this result geometrically. Verify the result for the special case } f(x) = \sin x.$$

5.7 Numerical Integration

To evaluate a definite integral $\int_a^b f(x) dx$ by means of the Fundamental Theorem of Calculus, it is necessary to find an antiderivative of f . If an antiderivative cannot be found, then numerical methods may be used to approximate the integral to any degree of accuracy. For example, if the norm of a partition of $[a, b]$ is small, then by Definition (5.8) the definite integral can be approximated by any Riemann sum of f . In particular, if we use a regular partition with $\Delta x = (b-a)/n$, then

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(w_i) \Delta x$$

where w_i is any number in the i th subinterval $[x_{i-1}, x_i]$ of the partition. Of course, the accuracy of the approximation depends upon the nature of f

and the magnitude of Δx . It may be necessary to make Δx very small in order to obtain the desired degree of accuracy. This, in turn, means that n is large, and hence the preceding sum contains many terms. Figure 5.2 illustrates the case in which $f(w_i)$ is the minimum value of f in $[x_{i-1}, x_i]$. In this case the error involved in the approximation is numerically the same as the area of the unshaded region that lies under the graph of f and over the inscribed rectangles.

If we take $w_i = x_{i-1}$, that is, if f is evaluated at the left-hand endpoint of each subinterval $[x_{i-1}, x_i]$, we may write

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(x_{i-1}) \Delta x.$$

If we let $w_i = x_i$, that is, if f is evaluated at the right-hand endpoint of $[x_{i-1}, x_i]$, then

$$\int_a^b f(x) dx \approx \sum_{i=1}^n f(x_i) \Delta x.$$

Another, and usually more accurate approximation, can be obtained by using the average of the last two approximations, that is,

$$\frac{1}{2} \left[\sum_{i=1}^n f(x_{i-1}) \Delta x + \sum_{i=1}^n f(x_i) \Delta x \right].$$

With the exception of $f(x_0)$ and $f(x_n)$, each $f(x_i)$ appears twice, and hence we may write this expression as

$$\frac{\Delta x}{2} \left[f(x_0) + \sum_{i=1}^{n-1} 2f(x_i) + f(x_n) \right].$$

Since $\Delta x = (b - a)/n$, this gives us the following rule.

Trapezoidal Rule (5.35)

If f is continuous on $[a, b]$ and if a regular partition of $[a, b]$ is determined by $a = x_0, x_1, \dots, x_n = b$, then

$$\int_a^b f(x) dx \approx \frac{b-a}{2n} [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + 2f(x_{n-1}) + f(x_n)].$$

The term *trapezoidal* arises from the case in which $f(x)$ is nonnegative on $[a, b]$. As illustrated in Figure 5.15, if P_k is the point with x -coordinate x_k on the graph of $y = f(x)$, then for each $i = 1, 2, \dots, n$, the points on the x -axis with x -coordinates x_{i-1} and x_i , together with P_{i-1} and P_i , are vertices of a trapezoid having area

$$\frac{\Delta x}{2} [f(x_{i-1}) + f(x_i)].$$

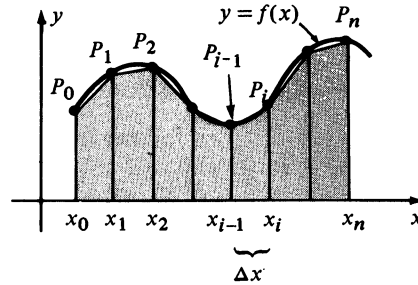


FIGURE 5.15

The sum of the areas of these trapezoids is the same as the sum in Rule (5.35). Hence, in geometric terms, the Trapezoidal Rule gives us an approximation to the area under the graph of f from a to b by means of trapezoids instead of the rectangles associated with Riemann sums.

The next result provides information about the maximum error that can occur if the Trapezoidal Rule is used to approximate a definite integral. The proof requires advanced methods and is therefore omitted.

**Error Estimate for
the Trapezoidal Rule (5.36)**

If M is a positive real number such that $|f''(x)| \leq M$ for all x in $[a, b]$, then the error involved in using the Trapezoidal Rule (5.35) is not greater than $M(b - a)^3/12n^2$.

Example 1 Approximate $\int_1^2 (1/x) dx$ by using the Trapezoidal Rule with $n = 10$. Estimate a maximum error in the approximation.

Solution Note that at this stage of our work it is impossible to find a function F such that $F'(x) = 1/x$ and hence we cannot use the Fundamental Theorem of Calculus to evaluate the integral. It is convenient to arrange our work as follows, where each $f(x_i)$ was obtained with a calculator and is accurate to four decimal places.

i	x_i	$f(x_i)$	m	$mf(x_i)$
0	1.0	1.0000	1	1.0000
1	1.1	0.9091	2	1.8182
2	1.2	0.8333	2	1.6666
3	1.3	0.7692	2	1.5384
4	1.4	0.7143	2	1.4286
5	1.5	0.6667	2	1.3334
6	1.6	0.6250	2	1.2500
7	1.7	0.5882	2	1.1764
8	1.8	0.5556	2	1.1112
9	1.9	0.5263	2	1.0526
10	2.0	0.5000	1	0.5000

The sum of the numbers in the last column is 13.8754. Since

$$(b - a)/2n = (2 - 1)/20 = 0.05,$$

it follows from (5.35) that

$$\int_1^2 \frac{1}{x} dx \approx (0.05)(13.8754) \approx 0.6938.$$

The error in the approximation may be estimated by means of (5.36). Since $f(x) = 1/x$ we have $f'(x) = -1/x^2$ and $f''(x) = 2/x^3$. The maximum value of $f''(x)$ on the interval $[1, 2]$ occurs at $x = 1$ and hence

$$|f''(x)| \leq \frac{2}{(1)^3} = 2.$$

Applying (5.36) with $M = 2$ we see that the maximum error is not greater than

$$\frac{2(2 - 1)^3}{12(10)^2} = \frac{1}{600} < 2 \times 10^{-3}. \quad \blacksquare$$

It will be shown in Chapter 7 that the integral in Example 1 equals the natural logarithm of 2, denoted by $\ln 2$. Using a calculator it is possible to verify that to five decimal places, $\ln 2$ is approximated by 0.69315. To obtain this approximation by means of the Trapezoidal Rule, it is necessary to use a very large value of n .

The following rule is usually more accurate than the Trapezoidal Rule.

Simpson's Rule (5.37)

Suppose f is continuous on $[a, b]$ and n is an even integer. If a regular partition is determined by $a = x_0, x_1, \dots, x_n = b$, then

$$\int_a^b f(x) dx \approx \frac{b - a}{3n} [f(x_0) + 4f(x_1) + 2f(x_2) + 4f(x_3) + \dots + 2f(x_{n-2}) + 4f(x_{n-1}) + f(x_n)].$$

The idea behind the proof of Simpson's Rule is that instead of using trapezoids to approximate the graph of f , we use portions of graphs of equations of the form $y = cx^2 + dx + e$. If $c \neq 0$, it can be shown that the graph of such an equation is a parabola. If $P_0(x_0, y_0)$, $P_1(x_1, y_1)$, and $P_2(x_2, y_2)$ are points, where $x_0 < x_1 < x_2$, then substituting the coordinates of P_0 , P_1 , and P_2 , respectively, into this equation produces three equations that may be solved for c , d , and e . As a special case, suppose h , y_0 , y_1 , and y_2 are positive, and consider the points $P_0(-h, y_0)$, $P_1(0, y_1)$, and $P_2(h, y_2)$, as illustrated in (i) of Figure 5.16.

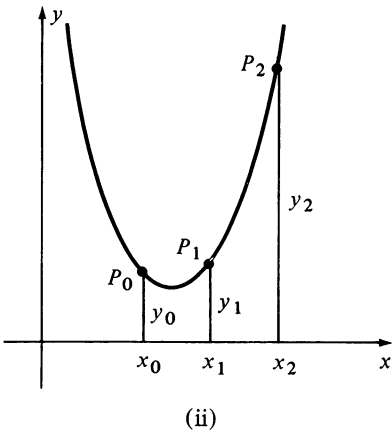
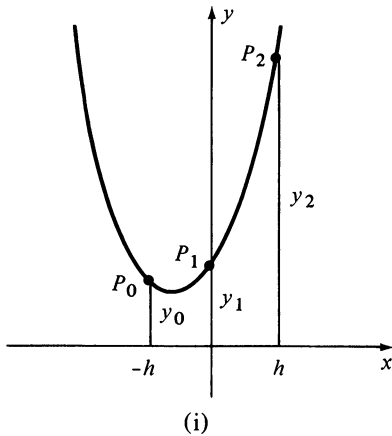


FIGURE 5.16

The area A under the graph of the equation from $-h$ to h is

$$A = \int_{-h}^h (cx^2 + dx + e) dx = \left[\frac{cx^3}{3} + \frac{dx^2}{2} + ex \right]_{-h}^h = \frac{h}{3}(2ch^2 + 6e).$$

Since the coordinates of P_0 , P_1 , and P_2 are solutions of $y = cx^2 + dx + e$, we have

$$y_0 = ch^2 - dh + e$$

$$y_1 = e$$

$$y_2 = ch^2 + dh + e$$

from which it follows that

$$y_0 + 4y_1 + y_2 = 2ch^2 + 6e.$$

Consequently,

$$A = \frac{h}{3}(y_0 + 4y_1 + y_2).$$

If the points P_0 , P_1 , and P_2 are translated horizontally, as illustrated in (ii) of Figure 5.16, then the area under the graph remains the same. Consequently, the preceding formula for A is true for *any* points P_0 , P_1 , and P_2 , provided $x_1 - x_0 = x_2 - x_1$.

If $f(x) \geq 0$ on $[a, b]$, then Simpson's Rule is obtained by regarding the definite integral as the area under the graph of f from a to b . Thus, suppose n is an even integer and $h = (b - a)/n$. We divide $[a, b]$ into n subintervals, each of length h , by choosing numbers $a = x_0, x_1, \dots, x_n = b$. Let $P_k(x_k, y_k)$ be the point on the graph of f with x -coordinate x_k , as illustrated in Figure 5.17.

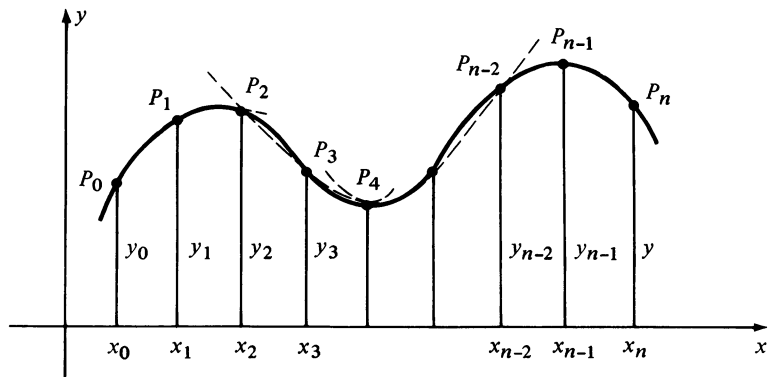


FIGURE 5.17

If the arc through P_0 , P_1 , and P_2 is approximated by the graph of an equation $y = cx^2 + dx + e$, then, as we have seen, the area under the graph of f from x_0 to x_2 is approximated by

$$\frac{h}{3}(y_0 + 4y_1 + y_2).$$

If the arc through P_2 , P_3 and P_4 is approximated in similar fashion, then the area under the graph of f from x_2 to x_4 is approximately

$$\frac{h}{3}(y_2 + 4y_3 + y_4).$$

We continue in this manner until we reach the last triple of points P_{n-2} , P_{n-1} , P_n and the corresponding approximation to the area under the graph, namely,

$$\frac{h}{3}(y_{n-2} + 4y_{n-1} + y_n).$$

Summing these approximations gives us

$$\int_a^b f(x) dx \approx \frac{h}{3}(y_0 + 4y_1 + 2y_2 + 4y_3 + \cdots + 2y_{n-2} + 4y_{n-1} + y_n)$$

which is the same as the sum in (5.37). If f is negative in $[a, b]$, then negatives of areas may be used to establish Simpson's Rule.

The following result may be established using advanced methods.

**Error Estimate for
Simpson's Rule (5.38)**

If M is a positive real number such that $|f^{(4)}(x)| \leq M$ for all x in $[a, b]$, then the error involved in using Simpson's Rule (5.37) is not greater than $M(b-a)^5/180n^4$.

Example 2 Approximate $\int_1^2 (1/x) dx$ by using Simpson's Rule with $n = 10$. Estimate the error in the approximation.

Solution This is the same integral considered in Example 1. We arrange our work as follows:

i	x_i	$f(x_i)$	m	$mf(x_i)$
0	1.0	1.0000	1	1.0000
1	1.1	0.9091	4	3.6364
2	1.2	0.8333	2	1.6666
3	1.3	0.7692	4	3.0768
4	1.4	0.7143	2	1.4286
5	1.5	0.6667	4	2.6668
6	1.6	0.6250	2	1.2500
7	1.7	0.5882	4	2.3528
8	1.8	0.5556	2	1.1112
9	1.9	0.5263	4	2.1052
10	2.0	0.5000	1	0.5000

The sum of the numbers in the last column is 20.7944. Since

$$\frac{b-a}{3n} = \frac{2-1}{30}$$

it follows from (5.37) that

$$\int_1^2 \frac{1}{x} dx \approx \left(\frac{1}{30}\right)(20.7944) \approx 0.6932.$$

We shall use (5.38) to estimate the error in the approximation. If $f(x) = 1/x$, the reader may verify that $f^{(4)}(x) = 24/x^5$. The maximum value of $f^{(4)}(x)$ on the interval $[1, 2]$ occurs at $x = 1$ and hence

$$|f^{(4)}(x)| \leq \frac{24}{(1)^5} = 24.$$

Applying (5.38) with $M = 24$, we see that the maximum error in the approximation is not greater than

$$\frac{24(2-1)^5}{180(10)^4} = \frac{2}{150000} < 1.4 \times 10^{-5}.$$

Note that this estimated error is much less than that obtained using the Trapezoidal Rule in Example 1. ■

At the beginning of this section we noted that a definite integral may be approximated by a Riemann sum $\sum_i f(w_i) \Delta x$. We then considered the special case where w_i was always a left-hand (or right-hand) endpoint of the i th subinterval $[x_{i-1}, x_i]$ of a regular partition of $[a, b]$. Another technique is to choose w_i as the *midpoint* of $[x_{i-1}, x_i]$. This leads to the next rule.

The Midpoint Rule (5.39)

If f is continuous on $[a, b]$ and if a regular partition of $[a, b]$ is determined by $a = x_0, x_1, \dots, x_n = b$, then

$$\int_a^b f(x) dx \approx \frac{b-a}{n} [f(\bar{x}_1) + f(\bar{x}_2) + \cdots + f(\bar{x}_n)]$$

where $\bar{x}_i = (x_{i-1} + x_i)/2$ is the midpoint of $[x_{i-1}, x_i]$.

The proof of (5.39) merely consists of noting that $\Delta x = (b-a)/n$ for each i . It can be shown that if M is a positive real number and $|f''(x)| \leq M$ for all x in $[a, b]$, then the error involved in using the Midpoint Rule is not greater than $M(b-a)^3/24n$.

Example 3 Approximate $\int_1^2 (1/x) dx$ by using the Midpoint Rule with $n = 10$.

Solution The integral is the same as that considered in the preceding two examples. Since the partition is determined by 1, 1.1, 1.2, $\dots$, 1.9, 2,

the midpoints of the subintervals are 1.05, 1.15, ..., 1.95, and the Midpoint Rule gives us

$$\int_1^2 \frac{1}{x} dx \approx 0.1 \left[\frac{1}{1.05} + \frac{1}{1.15} + \frac{1}{1.25} + \frac{1}{1.35} + \frac{1}{1.45} + \frac{1}{1.55} + \frac{1}{1.65} + \frac{1}{1.75} + \frac{1}{1.85} + \frac{1}{1.95} \right]$$

It is left to the reader to verify that this gives us the approximation $(0.1)(6.928) = 0.6928$. This number compares favorably with the values 0.6938 and 0.6932 obtained from the Trapezoidal Rule and Simpson's Rule. ■

An important aspect of numerical integration is that it can be used to approximate the integral of a function that is described in tabular form. To illustrate, suppose it is found experimentally that two physical variables x and y are related as shown in the following table.

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	3.1	4.0	4.2	3.8	2.9	2.8	2.7

If we regard y as a function of x , say $y = f(x)$, where f is continuous, then the definite integral $\int_1^4 f(x) dx$ may have an important physical meaning. In the present illustration, *the integral may be approximated without knowing the explicit formula for $f(x)$* . In particular, if the Trapezoidal Rule (5.35) is used, with $n = 6$ and $(b - a)/2n = (4 - 1)/12 = 0.25$, then

$$\begin{aligned} \int_1^4 f(x) dx &\approx 0.25[3.1 + 2(4.0) + 2(4.2) + 2(3.8) + 2(2.9) + 2(2.8) + 2.7] \\ &\approx 10.3. \end{aligned}$$

Since the number of subdivisions is even, we could also approximate the integral by means of Simpson's Rule.

5.7 Exercises

In Exercises 1–8 use (a) the Trapezoidal Rule (5.35) and (b) Simpson's Rule (5.37) to approximate the definite integral for the stated value of n . Use approximations to four decimal places for $f(x_i)$ and round off answers to two decimal places. (It is advisable to use a calculator for these exercises.)

1 $\int_1^4 \frac{1}{x} dx, n = 6$

2 $\int_0^3 \frac{1}{1+x} dx, n = 8$

3 $\int_0^1 \frac{1}{\sqrt{1+x^2}} dx, n = 4$

4 $\int_2^3 \sqrt{1+x^3} dx, n = 4$

5 $\int_0^2 \frac{1}{4+x^2} dx, n = 10$

6 $\int_0^{0.6} \frac{1}{\sqrt{4-x^2}} dx, n = 6$

7 $\int_0^\pi \sqrt{\sin x} dx, n = 6$

8 $\int_0^\pi \sin \sqrt{x} dx, n = 4$

9–12 Rework Exercises 1–4 using the Midpoint Rule (5.39).

13 Use the Trapezoidal Rule (5.35) with $(b - a)/n = 0.1$ to show that

$$\int_1^{2.7} \frac{1}{x} dx < 1 < \int_1^{2.8} \frac{1}{x} dx.$$

- 14 Find upper bounds for the errors in parts (a) and (b) of Exercise 1.

Suppose the tables in Exercises 15 and 16 were obtained experimentally, where x and y are physical variables. Assuming that $y = f(x)$ where f is continuous, approximate $\int_2^4 f(x) dx$ by means of (a) the Trapezoidal Rule; (b) Simpson's Rule; (c) the Midpoint Rule.

15

x	y
2.00	4.12
2.25	3.76
2.50	3.21
2.75	3.58
3.00	3.94
3.25	4.15
3.50	4.69
3.75	5.44
4.00	7.52

16

x	y
2.0	12.1
2.2	11.4
2.4	9.7
2.6	8.4
2.8	6.3
3.0	6.2
3.2	5.8
3.4	5.4
3.6	5.1
3.8	5.9
4.0	5.6

- 17 The graph in the accompanying figure was recorded by an instrument used to measure a physical quantity. Estimate y -coordinates of points on the graph and approximate the area of the shaded region by using (with $n = 6$) (a) the Trapezoidal Rule; (b) Simpson's Rule; (c) the Midpoint Rule.

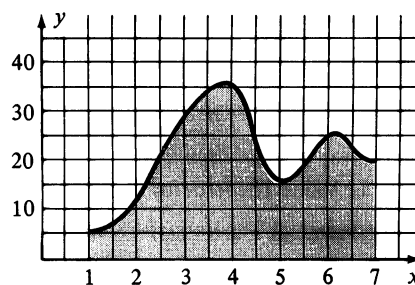


FIGURE FOR EXERCISE 17

- 18 A man-made lake has the shape illustrated in the accompanying figure, where adjacent measurements are 20 ft apart. Use the Trapezoidal Rule to estimate the surface area of the lake.

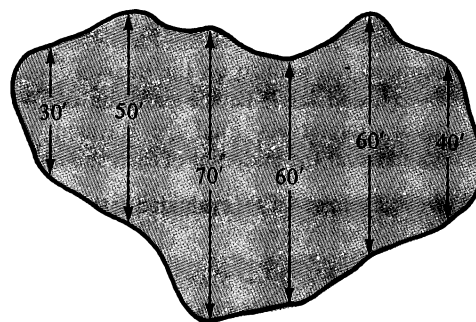


FIGURE FOR EXERCISE 18

- 19 Find the least integer n such that the error involved in approximating $\int_{1/2}^1 (1/x) dx$ is less than 0.0001 if (a) the Trapezoidal Rule is used; (b) Simpson's Rule is used.
- 20 Find the least integer n such that the error involved in approximating $\int_0^3 (1+x)^{-1} dx$ is less than 0.001 if (a) the Trapezoidal Rule is used; (b) Simpson's Rule is used.
- 21 If $f(x)$ is a polynomial of degree less than 4, prove that Simpson's Rule gives the exact value of $\int_a^b f(x) dx$.
- 22 If f is continuous and has nonnegative values, and if $f''(x) > 0$ throughout $[a, b]$, prove that $\int_a^b f(x) dx$ is less than the number given by the Trapezoidal Rule.

5.8 Review: Concepts

Define or discuss each of the following.

- The area under the graph of a nonnegative continuous function f from a to b
- Partition of $[a, b]$

- Norm of a partition

- Riemann sum

- $$\lim_{\|P\| \rightarrow 0} \sum_i f(w_i) \Delta x_i = I$$

- 6 The definite integral of f from a to b
- 7 Upper and lower limits of integration
- 8 Integrand
- 9 Properties of the definite integral
- 10 The Mean Value Theorem for Definite Integrals
- 11 The Fundamental Theorem of Calculus
- 12 Indefinite integral
- 13 Constant of integration
- 14 Power rule for indefinite integration
- 15 The method of substitution
- 16 Numerical integration

Review: Exercises

Determine the sums in Exercises 1–3.

- 1 $\sum_{k=1}^5 (k^2 + 3)$
- 2 $\sum_{k=1}^{50} 8$
- 3 $\sum_{k=0}^4 (-1/2)^{k-2}$
- 4 Suppose f is defined on the interval $[-2, 3]$ by $f(x) = 1 - x^2$, and let P be the partition of $[-2, 3]$ into five equal subintervals. Find the Riemann sum R_P if f is evaluated at the midpoint of each subinterval.
- 5 Work Exercise 4 if f is evaluated at the right-hand endpoint of each subinterval.
- 6 Given $\int_1^4 (x^2 + 2x - 5) dx$, find numbers that satisfy the conclusion of the Mean Value Theorem for Integrals.

Evaluate the integrals in Exercises 7–32.

- 7 $\int_0^1 \sqrt[3]{8x^7} dx$
- 8 $\int \sqrt[3]{5t+1} dt$
- 9 $\int_0^1 \frac{z^2}{(1+z^3)^2} dz$
- 10 $\int (x^2 + 4)^2 dx$
- 11 $\int (1 - 2x^2)^3 x dx$
- 12 $\int \frac{(1 + \sqrt{x})^2}{\sqrt[3]{x}} dx$
- 13 $\int_1^2 \frac{w+1}{\sqrt{w^2+2w}} dw$
- 14 $\int_1^2 \frac{s^2+2}{s^2} ds$
- 15 $\int \frac{1}{\sqrt{x}(1 + \sqrt{x})^2} dx$
- 16 $\int_1^2 \frac{x^2 - x - 6}{x+2} dx$
- 17 $\int (3 - 2x - 5x^3) dx$
- 18 $\int (y + y^{-1})^2 dy$
- 19 $\int_0^2 x^2 \sqrt{x^3 + 1} dx$
- 20 $\int_1^4 3x^2 \sqrt{x^3 + x} dx$
- 21 $\int (4t + 1)(4t^2 + 2t - 7)^2 dt$
- 22 $\int \frac{\sqrt[4]{1-v^{-1}}}{v^2} dv$
- 23 $\int (2x^{-3} - 3x^{-2}) dx$
- 24 $\int_1^9 \sqrt{2r+7} dr$
- 25 $\int \sin(3 - 5x) dx$
- 26 $\int x^2 \cos(2x^3) dx$
- 27 $\int \cos 3x \sin^4 3x dx$
- 28 $\int \frac{\sin(1/x)}{x^2} dx$
- 29 $\int_0^{\pi/2} \cos x \sqrt{3 + 5 \sin x} dx$
- 30 $\int \frac{\cos 3x}{\sin^3 3x} dx$

$$31 \quad \int_0^{\pi/4} \sin 2x \cos^2 2x \, dx$$

$$32 \quad \int_{\pi/6}^{\pi/4} (\sec x + \tan x)(1 - \sin x) \, dx$$

$$33 \quad \int D_y \sqrt[5]{y^4 + 2y^2 + 1} \, dy$$

$$34 \quad \int_0^{\pi/2} D_x (x \sin^3 x) \, dx$$

$$35 \quad D_x \int_0^1 (x^3 + x^2 - 7)^5 \, dx$$

$$36 \quad D_z \int_0^z (x^2 + 1)^{10} \, dx$$

37 Evaluate $\int_0^{10} \sqrt{1+x^4} \, dx$ by using (a) the Trapezoidal Rule with $n = 5$ and (b) Simpson's Rule with $n = 8$. Use approximations to four decimal places for $f(x_i)$ and round off answers to two decimal places.

38 If $f(x) = x^4 \sqrt{x^5 + 4}$, find the area of the region under the graph of f from 0 to 2.

39 Find the area of the region under the graph of $y = \sin(x/4)$ from $x = 0$ to $x = \pi$.

40 Use a definite integral to prove that the area of a right triangle of altitude a and base b is $ab/2$. (*Hint*: Take the vertices at the points $(0, 0)$, $(b, 0)$, and (b, a) .)

6 Applications of the Definite Integral

The definite integral is useful for solving a large variety of applied problems. In this chapter we shall discuss area, volume, work, liquid force, lengths of curves,

and some problems from economics and biology. Other applications will be considered later in the text.

6.1 Area

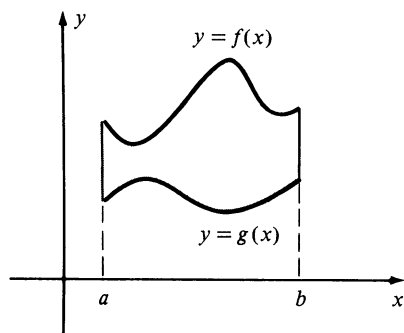


FIGURE 6.1

If a function f is continuous on a closed interval $[a, b]$, and $f(x) \geq 0$ for all x in $[a, b]$, then by Theorem (5.12) the area under the graph of f from a to b equals $\int_a^b f(x) dx$. If g is another continuous nonnegative valued function on $[a, b]$, and if $f(x) \geq g(x)$ for all x in $[a, b]$, then the area A of the region bounded by the graphs of f , g , $x = a$, and $x = b$ (see Figure 6.1) can be found by subtracting the area under the graph of g from the area under the graph of f , that is,

$$A = \int_a^b f(x) dx - \int_a^b g(x) dx = \int_a^b [f(x) - g(x)] dx.$$

This formula for A can be extended to the case in which f or g is negative for some x in $[a, b]$, as illustrated in (i) of Figure 6.2. To find the area of the indicated region, let us choose a negative number d less than the minimum value of g on $[a, b]$ and consider the two functions f_1 and g_1 defined by

$$f_1(x) = f(x) - d, \quad g_1(x) = g(x) - d$$

for all x in $[a, b]$. Since d is negative, values of f_1 and g_1 are found by adding the positive number $|d|$ to corresponding values of f and g , respectively. Geometrically, this amounts to raising the graphs of f and g a distance $|d|$,

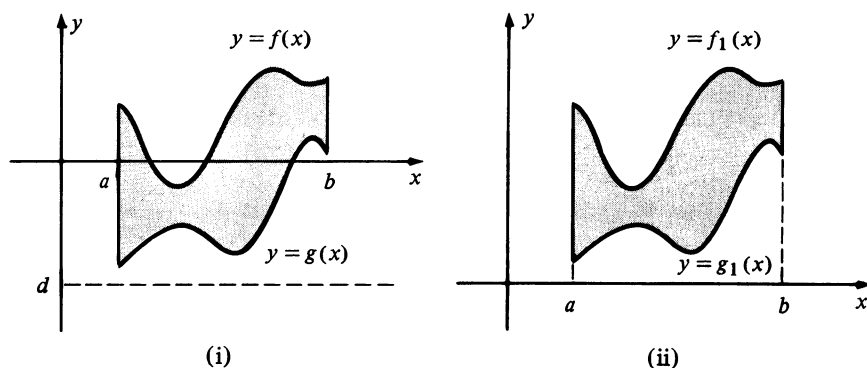


FIGURE 6.2

giving us a region having the same shape as the original region, but lying entirely above the x -axis.

If A is the area of the region in (ii) of Figure 6.2, then

$$\begin{aligned}
 A &= \int_a^b [f_1(x) - g_1(x)] dx \\
 &= \int_a^b \{[f(x) - d] - [g(x) - d]\} dx \\
 &= \int_a^b [f(x) - g(x)] dx.
 \end{aligned}$$

The preceding discussion may be summarized as follows.

Theorem (6.1)

If f and g are continuous and $f(x) \geq g(x)$ for all x in $[a, b]$, then the area A of the region bounded by the graphs of f , g , $x = a$, and $x = b$ is

$$A = \int_a^b [f(x) - g(x)] dx.$$

The formula for A in Theorem (6.1) may be interpreted as a limit of a sum. If we define the function h by $h(x) = f(x) - g(x)$, and if w is in $[a, b]$, then $h(w)$ is the distance from the point on the graph of g with x -coordinate w to the point on the graph of f with x -coordinate w . As in the discussion of Riemann sums in Chapter 5, let P denote a partition of $[a, b]$ determined by the numbers $a = x_0, x_1, \dots, x_n = b$. For each i , let $\Delta x_i = x_i - x_{i-1}$, and let w_i be an arbitrary number in the i th subinterval $[x_{i-1}, x_i]$ of P . By the definition of h ,

$$h(w_i) \Delta x_i = [f(w_i) - g(w_i)] \Delta x_i$$

which, in geometric terms, is the area of a rectangle of length $f(w_i) - g(w_i)$ and width Δx_i (see Figure 6.3).

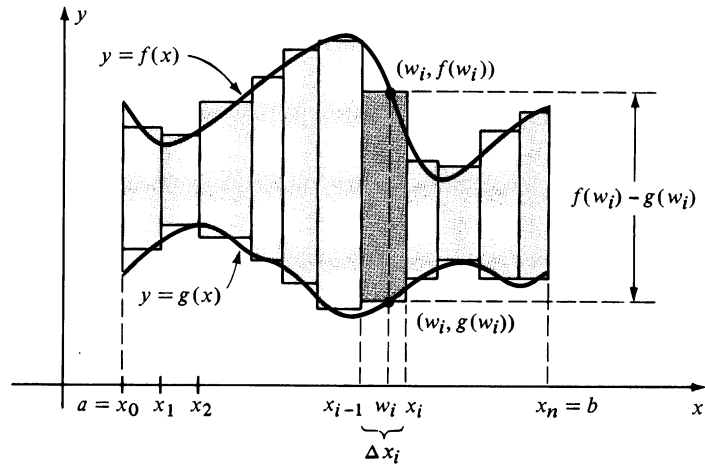


FIGURE 6.3

The Riemann sum

$$\sum_i h(w_i) \Delta x_i = \sum_i [f(w_i) - g(w_i)] \Delta x_i$$

is the sum of the areas of all rectangles pictured in Figure 6.3, and may be thought of as an approximation to the area of the region bounded by the graphs of f and g from $x = a$ to $x = b$. By the definition of the definite integral,

$$\lim_{\|P\| \rightarrow 0} \sum_i h(w_i) \Delta x_i = \int_a^b h(x) dx.$$

Using the definition of h gives us the following.

Theorem (6.2)

If f and g are continuous and $f(x) \geq g(x)$ for all x in $[a, b]$, then the area A of the region bounded by the graphs of f , g , $x = a$, and $x = b$ is

$$A = \lim_{\|P\| \rightarrow 0} \sum_i [f(w_i) - g(w_i)] \Delta x_i = \int_a^b [f(x) - g(x)] dx.$$

When we use Theorem (6.2) we initially think of approximating the region by means of rectangles of the type shown in Figure 6.3. After writing a formula for the area of a typical rectangle, we sum all such rectangles and, as in (6.2), take the limit of this sum to obtain the area of the region. This technique is illustrated in the following examples.

Example 1 Find the area of the region bounded by the graphs of the equations $y = x^2$ and $y = \sqrt{x}$.

Solution We shall employ the Riemann sum approach. The region and a typical rectangle are sketched in Figure 6.4.

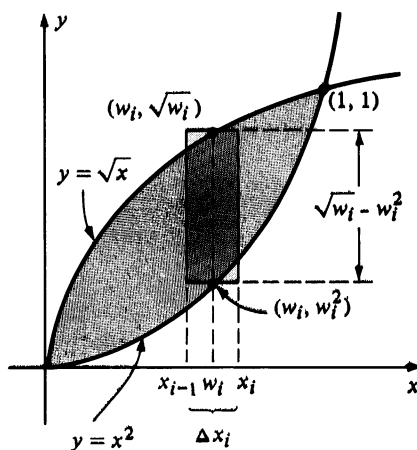


FIGURE 6.4

As indicated in the figure, the length of a typical rectangle is $\sqrt{w_i} - w_i^2$ and its area is $(\sqrt{w_i} - w_i^2) \Delta x_i$. Using Theorem (6.2) with $a = 0$ and $b = 1$, we obtain

$$\begin{aligned} A &= \lim_{\|P\| \rightarrow 0} \sum_i (\sqrt{w_i} - w_i^2) \Delta x_i \\ &= \int_0^1 (\sqrt{x} - x^2) dx \\ &= \left[\frac{2}{3} x^{3/2} - \frac{1}{3} x^3 \right]_0^1 = \frac{2}{3} - \frac{1}{3} = \frac{1}{3}. \end{aligned}$$

The area can also be found by direct substitution in Theorem (6.1), with $f(x) = \sqrt{x}$ and $g(x) = x^2$. ■

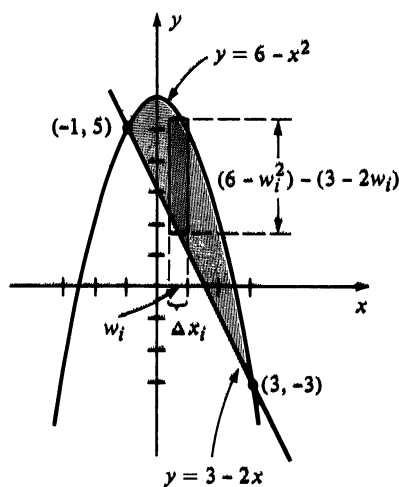


FIGURE 6.5

As illustrated in Example 1, to find areas by using limits of sums, we always begin by sketching the region with at least one typical rectangle, labeling the drawing appropriately. The reason for stressing the summation technique is that similar limiting processes will be employed later for calculating many other mathematical and physical quantities. Treating areas as limits of sums will make it easier to understand those future applications. At the same time it will help solidify the meaning of the definite integral.

Example 2 Find the area of the region bounded by the graphs of $y + x^2 = 6$ and $y + 2x - 3 = 0$.

Solution The region and a typical rectangle are sketched in Figure 6.5. The points of intersection $(-1, 5)$ and $(3, -3)$ of the two graphs may be

found by solving the two given equations simultaneously. To use Theorem (6.1) or (6.2) it is necessary to solve each equation for y in terms of x , obtaining

$$y = 6 - x^2 \quad \text{and} \quad y = 3 - 2x.$$

The functions f and g are then given by

$$f(x) = 6 - x^2 \quad \text{and} \quad g(x) = 3 - 2x.$$

As shown in Figure 6.5, the length of a typical rectangle is

$$(6 - w_i^2) - (3 - 2w_i)$$

where w_i is some number in the i th subinterval of a partition P of $[-1, 3]$. The area of this rectangle is

$$[(6 - w_i^2) - (3 - 2w_i)] \Delta x_i.$$

Applying Theorem (6.2),

$$\begin{aligned} A &= \lim_{\|P\| \rightarrow 0} \sum_i [(6 - w_i^2) - (3 - 2w_i)] \Delta x_i \\ &= \int_{-1}^3 [(6 - x^2) - (3 - 2x)] dx \\ &= \int_{-1}^3 (3 - x^2 + 2x) dx \\ &= \left[3x - \frac{x^3}{3} + x^2 \right]_{-1}^3 \\ &= \left[9 - \frac{27}{3} + 9 \right] - \left[-3 - \left(-\frac{1}{3} \right) + 1 \right] = \frac{32}{3}. \quad \blacksquare \end{aligned}$$

After students thoroughly understand the limit of a sum technique illustrated in Examples 1 and 2, it is customary to bypass the subscript part of the solution and proceed immediately to “setting up” the integral. To illustrate, in Example 2 we could regard dx as the width of a typical rectangle. The length of the rectangle is represented by the distance

$$(6 - x^2) - (3 - 2x)$$

between the upper and lower boundaries of the region, where x is an arbitrary number in $[-1, 3]$. Thus the area of a rectangle may be represented, in nonsubscript notation, by

$$[(6 - x^2) - (3 - 2x)] dx.$$

Placing the integral sign $\int_{-1}^3$ in front of this expression may be regarded as summing all such terms and simultaneously letting the widths of the rectangles approach 0.

Since one of our objectives is to emphasize the definition of the definite integral (5.8), we will continue to use limits of sums in many of the remaining examples in this chapter. The symbol w_i employed in solutions always

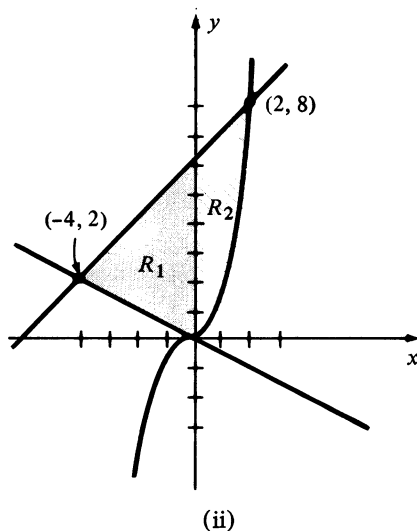
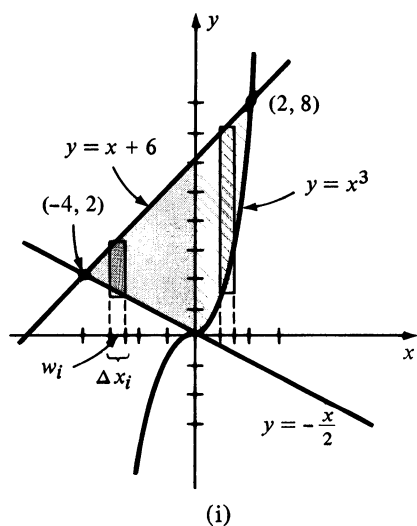


FIGURE 6.6

denotes a number in the i th subinterval of a partition. Those who wish to use the nonsubscript approach may (at the discretion of the instructor) proceed immediately to the definite integral which follows a stated limit of a sum.

The following example illustrates that it is sometimes necessary to subdivide a region and use Theorem (6.1) or (6.2) more than once in order to find the area.

Example 3 Find the area of the region R that is bounded by the graphs of $y - x = 6$, $y - x^3 = 0$, and $2y + x = 0$.

Solution The graphs and region are sketched in (i) of Figure 6.6 where, as indicated, each equation has been solved for y in terms of x so that appropriate functions may be introduced. Typical rectangles are shown extending from the lower boundary to the upper boundary of R . Since the lower boundary consists of portions of two different graphs, the area cannot be found by using only one definite integral. However, if R is divided into two subregions R_1 and R_2 , as shown in (ii) of Figure 6.6, then we can determine the area of each and add them together.

For region R_1 the upper and lower boundaries are the lines $y = x + 6$ and $y = -x/2$, respectively, and hence the length of a typical rectangle is

$$(w_i + 6) - (-w_i/2).$$

Applying Theorem (6.2), the area A_1 of R_1 is

$$\begin{aligned} A_1 &= \lim_{\|P\| \rightarrow 0} \sum_i \left[(w_i + 6) - \left(-\frac{w_i}{2} \right) \right] \Delta x_i \\ &= \int_{-4}^0 \left[(x + 6) - \left(-\frac{x}{2} \right) \right] dx \\ &= \int_{-4}^0 \left[\frac{3}{2}x + 6 \right] dx = \left[\frac{3}{4}x^2 + 6x \right]_{-4}^0 \\ &= 0 - (12 - 24) = 12. \end{aligned}$$

Region R_2 has the same upper boundary $y = x + 6$ as R_1 ; however, the lower boundary is given by $y = x^3$. In this case the length of a typical rectangle is

$$(w_i + 6) - w_i^3$$

and the area A_2 of R_2 is

$$\begin{aligned} A_2 &= \lim_{\|P\| \rightarrow 0} \sum_i [(w_i + 6) - w_i^3] \Delta x_i \\ &= \int_0^2 [(x + 6) - x^3] dx \\ &= \left[\frac{x^2}{2} + 6x - \frac{x^4}{4} \right]_0^2 \\ &= (2 + 12 - 4) - 0 = 10. \end{aligned}$$

The area of the entire region R is, therefore, $A_1 + A_2 = 12 + 10 = 22$. ■